

Every complex K3 surface is Oka

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Abstract

We prove that every complex K3 surface X is Oka, resolving the K3 Oka conjecture. Equivalently, for every $m \geq 1$, every holomorphic map to X from a neighborhood of a compact convex set in $\mathbb{C}^m$ can be approximated uniformly on that set by entire maps $\mathbb{C}^m \rightarrow X$.

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1 Introduction

A *complex K3 surface* is a compact connected complex surface that is simply connected and has trivial canonical bundle. Thus it carries a nowhere-zero holomorphic two-form. Every complex K3 surface is Kähler, but it need not be projective or contain any curve [24, Chapters 1 and 7]. This combination of a rigid cohomological structure and potentially sparse algebraic geometry makes K3 surfaces a natural setting in which to study holomorphic flexibility.

The *convex approximation property* (CAP) for a complex manifold asks that maps holomorphic near a compact convex subset of $\mathbb{C}^m$ can be approximated there by maps defined on all of $\mathbb{C}^m$, for every source dimension m . Forstnerič proved that CAP characterizes Oka manifolds [13, 14]. In its equivalent Stein-source formulation, the Oka property includes deformation of continuous maps to holomorphic maps, with approximation and holomorphic interpolation. Our main result is the following precise approximation statement.

Theorem 1.1. *Let X be a complex K3 surface, with any smooth Hermitian distance d_X . For every integer $m \geq 1$, every nonempty compact convex set $K \subset \mathbb{C}^m$, every open neighborhood U of K , every holomorphic map $f : U \rightarrow X$, and every $\epsilon > 0$, there exists a holomorphic map $F : \mathbb{C}^m \rightarrow X$ such that*

$$\sup_{z \in K} d_X(F(z), f(z)) < \epsilon.$$

Theorem 1.1 answers the K3 question in [15, Section 8, Problem C] and proves the positive conjecture formulated explicitly in [30, Introduction]. Its scope includes nonprojective surfaces and imposes no condition on the Picard rank or existence of an elliptic fibration.

An *entire curve* is a holomorphic map from $\mathbb{C}$, and it is an *immersion* if its derivative never vanishes. The approximation and interpolation theory gives the following concrete consequence, proved in Section 10.

Corollary 1.2. *Let S be a complex K3 surface. For every $x \in S$ and every nonzero vector $v \in T_x S$, there is a holomorphic immersion $f : \mathbb{C} \rightarrow S$ such that*

$$f(0) = x, \quad f'(0) = v, \quad \overline{f(\mathbb{C})} = S,$$

where closure is taken in the ordinary complex topology. If S is projective, then $f(\mathbb{C})$ is Zariski dense.

Here the tangent vector, not just its line, is prescribed. For projective S , the density conclusion confirms the K3 case of Campana’s prediction of Zariski-dense entire curves on special projective

manifolds [7]. Here specialness means the absence of a meromorphic fibration whose positive-dimensional base, endowed with the orbifold structure induced by the multiple fibers, is of general type. See [6, Conjecture 1.3] for the prediction and [6, Proposition 8.9] for the specialness of projective varieties with trivial canonical bundle. The ordinary density and prescribed first jet are additional conclusions.

Section 10.2 derives the Oka property for Enriques surfaces and hence for all minimal compact complex surfaces of Kodaira dimension zero. It also obtains global dominating sprays on projective K3 and Enriques surfaces. A separate class-VII classification there combines earlier Oka results with the global spherical shell theorem of the companion manuscript [26]; that theorem is not an input to the K3 argument.

1.1 Earlier results and the remaining obstacles

The use of holomorphic sprays and approximation to realize topological flexibility is central to Gromov’s Oka principle for elliptic bundles [21]. Forstnerič’s convex approximation characterization [13, Theorem 0.1] makes CAP an effective criterion for this flexibility, while his interpolation theorem [12, Corollary 1.3] links it to prescribed holomorphic data on subvarieties. Our proof uses local chart sprays and splitting arguments in this tradition, with the uniform estimates needed for the strip and convex-face constructions made explicit.

Holomorphic flexibility on K3 surfaces has been studied through entire curves, domination by affine space, and approximation. Buzzard and Lu proved that elliptic and Kummer K3 surfaces are holomorphically dominable by $\mathbb{C}^2$ [5, Propositions 4.4–4.5]. Dominability means the existence of a map $\mathbb{C}^2 \rightarrow X$ whose differential is surjective somewhere. Forstnerič and Lárusson strengthened this for every Kummer surface to *strong dominability*: such a map can pass through each prescribed target point with surjective differential [16, Corollary 3]. These properties concern individual maps; CAP asks for approximation of every local map in every source dimension.

For arbitrary complex K3 surfaces, Kamenova, Lu and Verbitsky proved that the Kobayashi pseudodistance vanishes [25, Corollary 2.2]; the period-dynamics argument is corrected in [29, Section 4]. This removes an intrinsic obstruction to holomorphic flexibility, but does not provide approximation or interpolation.

Alarcón and Forstnerič introduced the Oka-1 property and established it for Kummer surfaces and elliptic K3 surfaces [1, Proposition 8.4 and Corollary 8.6]. Oka-1 concerns approximation and discrete jet interpolation from open Riemann surfaces, within prescribed continuous homotopy classes. Its full formulation is recalled below. The source dimension distinction matters: the one-dimensional property alone does not give Theorem 1.1.

For projective K3 geometry, Chen, Gounelas and Liedtke developed existence results for curves of prescribed geometric genus [10]. Chen and Gounelas then proved that every projective K3 surface contains genus-one curves of unbounded self-intersection whose smooth normalizations vary in moduli [9, Theorem A]. We use two of their families with different ample degrees. Their complete genus-one fibers supply holomorphic flows for all complex times, and the difference in degrees forces their tangent directions to be generically independent.

More recently, Xie and Zhao proved that smooth $(2, 2, 2)$ hypersurfaces in $(\mathbb{P}^1)^3$ are Oka and established dense G_δ sets of Oka K3 periods and corresponding local deformation parameters [30, Theorems A, B, and E]. Their work combines geometric constructions of flexibility with arithmetic dynamics of periods and Torelli theory. We use the same period-theoretic framework, including the corrected orbit alternatives of Verbitsky [29], with different geometric constructions. A dense collection of Oka surfaces does not by itself settle the property for each surface: the transfer must

address every possible rational subspace of its period plane.

Two analytic obstacles arise even before that transfer. Local approximation in one variable must retain arbitrary holomorphic parameters and then imply approximation on arbitrary convex sets. Moreover, the geometric constructions produce long chains of annuli; errors at their joins must be corrected without exhausting a fixed positive transverse width. We handle the first issue by a polynomial change of source coordinates and gluing estimates measured on real faces. We handle the second by exact symplectic sewing with bounds independent of the chain lengths.

1.2 The proof and its organization

The analytic and geometric parts meet at a local operation, denoted L . It enlarges a planar disc across an attached disc, for maps with image in a sufficiently small target neighborhood, while retaining any prescribed smaller polydisc of passive holomorphic parameters. Section 2 defines L precisely and obtains it from two strip maps, one defined on $\mathbb{C} \times \Delta$ and the other on $\Delta \times \mathbb{C}$, with transverse complete directions. Aligning their coordinate axes exactly makes the overlap error arbitrarily small. A bounded additive splitting and a contraction correct it. The same section propagates L across the center of a small disc whose boundary already lies in the locus where L holds.

Sections 3 and 4 prove that L at every point implies CAP in every dimension (Theorem 3.1). A polynomial automorphism concentrates the part of an enlarged polydisc outside the original domain into one long coordinate and bounded transverse coordinates. The planar operation with passive parameters then gives approximation on polydiscs. To pass to an arbitrary convex set, we remove the defining real faces of a containing polytope one at a time. The necessary gluing estimate uses Hölder accuracy on each real face, without requiring a fixed complex thickness on which that accuracy holds.

Section 5 constructs the two complete directions on projective K3 surfaces using the Chen–Gounelas families. Propagation across their algebraic exceptional set gives L everywhere. This handles one case of the final period argument and also exhibits a direct source of the required local geometry.

The nonprojective cases require strips on open regions of period space. Section 6 first establishes the uniform sewing theorem. On an annulus, a closed holomorphic one-form can have a nonzero period. Exactness of the symplectic transition removes this obstruction: the transverse constant Fourier coefficient becomes quadratic in the error. The other Fourier modes admit a splitting whose bounds are independent of the number and lengths of the annuli. A quadratic iteration then retains a positive transverse radius along an infinite chain.

To describe the geometric transfer, fix the K3 lattice Λ , the integral second-cohomology lattice with its intersection form. A *marking* identifies $H^2(X, \mathbb{Z})$ with Λ . The real and imaginary parts of a nonzero holomorphic two-form span an oriented positive two-plane, the *period* of the marked surface. Its rational directions are the vectors in its intersection with $\Lambda \otimes \mathbb{Q}$.

Section 7 constructs a nonempty open region in the full period domain on which every surface has L everywhere. It smooths a union of two quadrics and sews annuli along their common elliptic curve. The chart estimates persist under unrestricted small K3 deformations. Section 8 constructs a second kind of region: for each fixed positive integral vector v , a relatively open set among the period planes containing v . Here an isotrivial genus-one fibration supplies the central model. Conservation of imaginary symplectic action keeps all recentered annuli away from singular fibers. Two independent cycles supply distinct strip leaves after deformation and hence the transverse directions needed for L .

Finally, Section 9 applies Ratner’s orbit theorem and Verbitsky’s corrected rational-direction analysis. Periods with no rational direction have orbits meeting the first open region. Periods with exactly one rational direction have orbits, under its stabilizer, meeting the corresponding second region. Rational period planes give projective K3 surfaces, already treated above. Torelli theory transfers L along these orbits, and Theorem 3.1 finishes the proof. Section 10 derives interpolation, Corollary 1.2, and the further surface consequences.

Conventions and standard inputs. Maps on a compact set are always defined on a neighborhood of that set. Approximation may lose any prescribed positive amount of an auxiliary domain margin. A strip means $\mathbb{C} \times \Delta_b$, with no injectivity requirement unless stated; local regularity is specified at each use. The analytic background consists of Stein theory, scalar approximation and the $L^2 \bar{\partial}$ theorem [14, 23, 28]. We recall the needed K3 period, Torelli, and nef-pencil results at their applications [24, Chapters 2, 6–8, and 11]. The arithmetic inputs are likewise stated where used. The local approximation, sewing, geometric constructions, and reductions between them are proved here.

2 Local approximation from complete directions

The immediate goal is an operation that enlarges one planar source variable while retaining arbitrary passive holomorphic parameters. Two transverse complete strip maps provide this operation; a second construction propagates it across points not initially covered by strips. These are the inputs to the convex approximation argument of Section 4.

Write $\Delta_r = \{z \in \mathbb{C} : |z| < r\}$. A map on a compact set will always mean a map holomorphic on an open neighborhood of that set. A *special pair* $D \subset D'$ consists of two closed topological discs with piecewise smooth boundary, where D' is obtained from D by attaching another such disc along a proper boundary arc. All approximation assertions use an arbitrary fixed Hermitian distance on the target.

Definition 2.1. A complex surface X has property L at x if there is an open neighborhood V of x with the following property. For every special pair $D \subset D'$, every pair of closed polydiscs $P_0 \Subset \text{int } P \subset \mathbb{C}^q$, and every holomorphic map f from a neighborhood of $D \times P$ into V , the restriction $f|_{D \times P_0}$ is a uniform limit of maps holomorphic near $D' \times P_0$ with values in X . The case $q = 0$ is included. We call V a *witness neighborhood*.

Witness neighborhoods can be shrunk, and the locus where L holds is consequently open.

2.1 Planar splitting and two strip maps

In the first strip the first coordinate is complete; in the second strip the second coordinate is complete. Their agreement below is along a germ of the first strip’s complete central curve and a transverse starting section of the second strip, not along both complete central curves. This agreement will make the overlap error small without restricting either complete direction. The conclusion is also allowed at a displaced point on the second strip’s central curve, a feature needed later when two leaves meet tangentially.

Proposition 2.2 (Two complete directions). *Suppose there are holomorphic maps*

$$\sigma_1 : \mathbb{C} \times \Delta_b \longrightarrow X, \quad \sigma_2 : \Delta_b \times \mathbb{C} \longrightarrow X$$

which are locally biholomorphic at $(0, 0)$ and satisfy $\sigma_1(z, 0) = \sigma_2(z, 0)$ for small z . If σ_2 is locally biholomorphic at $(0, c)$, then L holds at $x = \sigma_2(0, c)$.

The same conclusion holds for a sequence of pairs whose long coordinate domains are discs of radii tending to infinity, whose bounded-coordinate radii are bounded below, and whose maps converge near $(0, 0)$ and, for the second map, near $(0, c)$ to the above nondegenerate germs. Exact axis agreement is required for every pair. In this version x is the value of the limiting second germ at $(0, c)$.

To construct an extension, we will choose a polynomial map $h = (h_1, h_2)$ on the source. On one planar piece only h_2 must be small, so that the first strip can be used; on the complementary pieces only h_1 must be small, so that the second strip can be used. Both coordinates are small on their overlaps. The next lemma arranges these pieces while keeping the original disc in one of the second-strip pieces.

Lemma 2.3. *Given a special pair $D \subset D'$ and an open neighborhood N of D , there are compact sets $A, B_0, B_1 \subset \mathbb{C}$ and bounded open sets U_A, U_0, U_1 such that*

$$\begin{aligned} \overline{U_A} &\subset \text{int } A, & \overline{U_i} &\subset \text{int } B_i \quad (i = 0, 1), \\ D &\subset U_0, & D' &\subset U_A \cup U_0 \cup U_1, \\ A \cap D &= \emptyset, & B_0 \cap B_1 &= \emptyset, & B_0 &\subset N. \end{aligned}$$

The compact sets $D \cup A$ and $B_0 \cup B_1$ are polynomially convex. For $U_B = U_0 \cup U_1$, the two closed fringes $\overline{U_A} \setminus \overline{U_B}$ and $\overline{U_B} \setminus \overline{U_A}$ have positive distance.

Proof. An ambient homeomorphism of the plane takes the attached-disc pair to $D = [-2, 0] \times [-1, 1]$ and $D' = [-2, 3] \times [-1, 1]$. Indeed, compatible parametrizations of the attaching arc and the remaining boundary arcs extend over both discs by Jordan–Schoenflies; the resulting outer boundary parametrization extends across the exterior. In this model choose

$$0 < c < c' < a' < a < b < b' < d' < d < 3, \quad 0 < \delta < \epsilon.$$

Use the following horizontal intervals:

	compact piece	open piece
B_0, U_0	$[-2 - \epsilon, a]$	$(-2 - \delta, a')$
A, U_A	$[c, d]$	(c', d')
B_1, U_1	$[b, 3 + \epsilon]$	$(b', 3 + \delta)$.

All compact rectangles have vertical interval $[-1 - \epsilon, 1 + \epsilon]$, and all open rectangles have interval $(-1 - \delta, 1 + \delta)$. Take a, ϵ small enough that B_0 is in the prescribed neighborhood. The closed fringes are disjoint compact sets, as is seen from their horizontal intervals $[a', b']$ and $[-2 - \delta, c'] \cup [d', 3 + \delta]$. Pull the sets back by the homeomorphism. Compact containment and positive separation persist. The compact unions in the statement are unions of two disjoint filled Jordan discs, hence have connected complement and are polynomially convex. $\square$

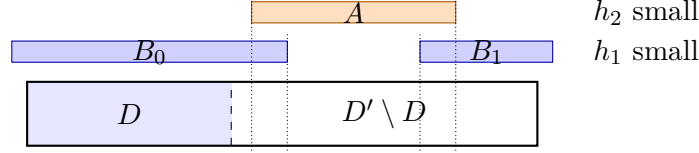


Figure 1: Horizontal extents of the three compact pieces, shown in separate rows. In the rectangular model, all three pieces share a vertical interval slightly larger than that of D' . The two overlap bands control both coordinates.

Figure 1 displays the complementary coordinate control for $h = (h_1, h_2)$: the first coordinate is small on the two outer pieces, and the second is small on the middle piece.

The following is the additive Cousin splitting underlying holomorphic spray gluing; compare [11, Lemma 4.6]. We give its integral construction to retain bounds independent of the number and radii of the passive parameters.

Lemma 2.4. *Let $U_A, U_B \subset \mathbb{C}$ be bounded open sets with positively separated closed fringes, and put $C = U_A \cap U_B$. For every open parameter polydisc P and every k , there are bounded linear operators*

$$\begin{aligned} R_A : H^\infty(C \times P, \mathbb{C}^k) &\longrightarrow H^\infty(U_A \times P, \mathbb{C}^k), \\ R_B : H^\infty(C \times P, \mathbb{C}^k) &\longrightarrow H^\infty(U_B \times P, \mathbb{C}^k) \end{aligned}$$

satisfying $R_A u - R_B u = u$. Their bounds depend only on the planar sets and not on P or its dimension.

Proof. Choose a smooth function χ on the plane, zero near $\overline{U_A} \setminus U_B$ and one near $\overline{U_B} \setminus U_A$, with bounded first derivative. On U_A extend χu by zero off C ; on U_B extend $(\chi - 1)u$ similarly. Both extensions have planar $\bar{\partial}$ derivative $u \bar{\partial} \chi$ on C and zero elsewhere in $\Omega = U_A \cup U_B$. Extend this bounded coefficient by zero to $\mathbb{C}$ and set

$$v(s, t) = \frac{1}{\pi} \int_C \frac{u(\zeta, t) \partial_{\bar{\zeta}} \chi(\zeta)}{s - \zeta} dA(\zeta).$$

The distributional identity $\partial_{\bar{s}}(1/(\pi s)) = \delta_0$ gives the required $\bar{\partial}$ equation. If d is the diameter of Ω , then

$$\|v\|_{\Omega \times P} \leq 2d \|\bar{\partial} \chi\|_{\infty} \|u\|_{C \times P}.$$

The kernel is locally integrable, so v is continuous in s and holomorphic in t . Thus $R_A u = \chi u - v$ and $R_B u = (\chi - 1)u - v$ are holomorphic, have the asserted bounds, and differ by u . No boundary continuity of u is needed. $\square$

Proof of Proposition 2.2. First consider complete strips. Choose $\rho > 0$ so small that $5\rho < b$, that $\tau = \sigma_2^{-1} \circ \sigma_1$ is defined near $\bar{\Delta}_{3\rho}^2$ using the inverse branch at the origin, and that $D\tau$ is invertible there. Exact axis agreement gives $\tau(z, 0) = (z, 0)$. The matrices $T_0(z) = D\tau(z, 0)$ and their inverses are bounded on $|z| \leq 2\rho$. Choose a biholomorphic σ_2 chart Q_c about $(0, c)$ whose first coordinate has modulus less than $\rho/16$, and take a relatively compact smaller image V as the witness neighborhood.

Fix input data as in Definition 2.1. Choose an open polydisc P_* with $P_0 \Subset P_* \Subset \text{int } P$. In the chart Q_c write $g = \sigma_2^{-1} \circ f = (g_1, g_2)$. There is a planar neighborhood N of D where g is

defined for parameters in a neighborhood of $\bar{P}_*$ and $|g_1| < \rho/8$. No smallness of $g_2 - c$ beyond membership in Q_c is used in the overlap calculation.

Choose the pieces of Lemma 2.3. Products of polynomially convex compact sets are polynomially convex. Scalar Oka–Weil approximation therefore gives a polynomial h_1 in the planar and parameter variables approximating g_1 on $B_0 \times \bar{P}_*$ and zero on $B_1 \times \bar{P}_*$. Its error ϵ_1 can be arbitrarily small; in particular, $|h_1| < \rho/2$ on their union. Independently, for every sufficiently small $\eta > 0$ there is a polynomial h_2 satisfying

$$\|h_2 - g_2\|_{D \times \bar{P}_*} < \eta, \quad \|h_2\|_{A \times \bar{P}_*} < \eta.$$

Let $h = (h_1, h_2)$ and $C = U_A \cap U_B$. On $U_B \times P_*$, $T = T_0(h_1)$ is holomorphic and boundedly invertible. We seek corrections satisfying

$$\tau(h + \alpha) = h + \beta, \quad \alpha = R_A u, \quad \beta = T R_B u.$$

By Lemma 2.4, this is the fixed-point equation

$$u = \mathcal{T}(u) := R_A u + T^{-1}(h - \tau(h + R_A u))$$

in $H^\infty(C \times P_*, \mathbb{C}^2)$.

On the overlap $|h_1| < \rho/2$ and $|h_2| < \eta$. Bounded first and second derivatives of τ , and its exact axis identity, give a constant $M \geq 1$ with

$$\|T\|, \|T^{-1}\| \leq M, \quad \|\tau(h) - h\| \leq M\eta, \quad \|D\tau(h + \alpha) - T\| \leq M(\eta + \|\alpha\|).$$

These constants do not depend on the degrees or uncontrolled coordinates of the polynomials. If $\kappa \geq 1$ bounds both splitting operators and $L_0 = 2M^2 + 1$, then on $\|u\| \leq L_0\eta$,

$$\|\mathcal{T}(0)\| \leq M^2\eta, \quad \|D\mathcal{T}(u)\| \leq M^2\kappa(1 + \kappa L_0)\eta.$$

For small η the last bound is at most $1/2$, the ball maps strictly into itself, and all transition evaluations remain in $\Delta_{3\rho}^2$. The contraction theorem gives

$$\|\alpha\| \leq \kappa L_0\eta, \quad \|\beta\| \leq M\kappa L_0\eta.$$

On $U_A \times P_*$ only the second coordinate of $h + \alpha$ has to be small, and on $U_B \times P_*$ only the first coordinate of $h + \beta$ has to be small. Consequently $\sigma_1(h + \alpha)$ and $\sigma_2(h + \beta)$ are defined on these full domains and agree on their overlap. On $D \times P_*$, the second formula is evaluated close to g in Q_c and approximates f as $\epsilon_1, \eta \rightarrow 0$. Since the union of the planar domains contains D' and $P_0 \Subset P_*$, this proves L .

For the finite-length version choose V , g , and the two polynomials using the limiting chart at $(0, c)$. Convergence on slightly larger compact coordinate neighborhoods gives uniform first and second derivative bounds, uniform inverse-chart domains, and thus uniform constants in the contraction argument. Exact axis agreement gives the same $O(\eta)$ initial error for every pair. The polynomials are bounded on the fixed bounded planar pieces and $\bar{P}_*$, so only a finite range of either long coordinate is used. Choose a sufficiently late pair to contain these ranges and make its chart near $(0, c)$ sufficiently close to the limiting chart. The construction then approximates the original data with the requested accuracy. $\square$

Remark 2.5. Two strip maps locally biholomorphic at chosen meeting preimages, with transverse central curves there, can be put in the form of Proposition 2.2. Locally lift the first central curve through the inverse of the second strip, writing the lift as $(a(z), t(z))$, where its bounded coordinate is a . Replace the second strip by $\tilde{\sigma}_2(z, w) = \sigma_2(a(z), t(z) + w)$. Transversality says $a'(0) \neq 0$, and time zero gives exact axis agreement. If the original second central curve also passes through x and its strip is regular there, some finite c gives a regular chart $\tilde{\sigma}_2(0, c) = x$. For finite-length families whose central germs converge to transverse germs through a common point, the implicit function theorem first gives nearby intersection points and meeting preimages. Translate the preimages to the origin and apply the same construction. Convergence and transversality give a uniformly positive bounded-coordinate radius; the translations and the function t lose only a bounded amount of long-coordinate length. Thus this construction also supplies the aligned families in the finite-length part of the proposition.

2.2 Sprays, deformation, and planar gluing

The strip criterion provides approximation when all input values lie in one witness neighborhood. To apply it successively to a map whose image meets several such neighborhoods, we need coordinates around the graph of the map and a way to join close maps in those coordinates. The next three lemmas supply exactly these tools. The use of sprays and their gluing follows the framework of Gromov [21] and Forstnerič [14]; here the sprays are constructed only near the graph on a relatively compact Stein source.

Lemma 2.6 (Local chart sprays). *Let $f : S \rightarrow X$ be holomorphic on a Stein manifold, let $K \Subset S$ be compact, and assume f^*TX is holomorphically trivial on S . After shrinking the source neighborhood there is a holomorphic map $f^\sharp(z, v)$ for z near K and $v \in \Delta_r^2$, with $f^\sharp(z, 0) = f(z)$, such that $v \mapsto f^\sharp(z, v)$ is injective with invertible differential. The hypothesis on f^*TX holds for a contractible Stein source and for a planar domain times a polydisc.*

Proof. The graph of f is a closed Stein submanifold of $S \times X$ and has a Stein neighborhood by the Stein neighborhood theorem [28]. On that neighborhood consider the vertical tangent bundle for projection to S . A holomorphic frame along the graph extends to holomorphic vertical vector fields by Cartan's Theorem B. Compose their small-time flows. The resulting map has the prescribed value and a frame as its vertical differential at $v = 0$. Compactness and the holomorphic inverse function theorem give a common radius on a neighborhood of K and injectivity in each parameter fiber. For the last assertion, a planar domain has the homotopy type of a one-dimensional CW complex, so every complex vector bundle on its product with a polydisc is topologically trivial. The Oka–Grauert principle makes it holomorphically trivial. The same argument applies to a contractible Stein source [14]. $\square$

Lemma 2.7 (Deforming a map on a relatively compact Stein domain). *Let $\pi : \mathcal{X} \rightarrow B$ be a holomorphic family, and let $f : S \rightarrow \mathcal{X}_{b_0}$ be holomorphic from a Stein manifold with image in the submersion locus of π . On every relatively compact source domain there is a holomorphic family of maps f_b into $\mathcal{X}_b$ for b near b_0 , with $f_{b_0} = f$.*

Proof. Work in local coordinates on B at b_0 . The graph of f has a Stein neighborhood in S times the submersion locus. On this neighborhood the surjection from tangent vectors vertical over S to the pulled-back base tangent bundle admits holomorphic lifts of the base coordinate vector fields: its kernel is coherent, and Cartan's Theorem B gives surjectivity on global sections. Flow the lifts successively for the small coordinate displacements of b . On a relatively compact source set a

common time radius is available. The flows are vertical over S , and their projections to B are the desired coordinate translations. Their composition is the required family of maps. $\square$

Lemma 2.8 (Planar spray gluing). *Suppose two holomorphic sprays over bounded planar domains U_A, U_B and a parameter polydisc are uniformly close on their overlap, with a positive margin in an auxiliary spray polydisc. Suppose the first spray is a chart spray there with uniform inverse-chart margins, and the closed planar fringes are positively separated. If the closeness is sufficiently small, their central maps can be glued after arbitrarily small holomorphic corrections in the auxiliary variables. The glued map is arbitrarily close to the first central map on any prescribed compact subset of its domain on which that spray is fixed, or has a uniform modulus of continuity in its auxiliary variables.*

Proof. Write the sprays as F_A and F_B . Compact chart margins give the transition $\gamma(s, t, v) = F_A(s, t, \cdot)^{-1}(F_B(s, t, v)) = v + E(s, t, v)$ on the overlap, with both $\|E\|$ and $\|D_v E\|$ small on a smaller auxiliary polydisc. The derivative estimate follows from the original closeness by Cauchy estimates on the larger polydisc. Equality of the resulting maps asks for $a = \gamma(b)$. Use $a = R_A u$, $b = R_B u$. Then

$$u = E(s, t, R_B u)$$

is a contraction on a small ball in the overlap H^∞ space. Lemma 2.4 supplies its uniform bounds and preserves holomorphic dependence on the passive parameters. The resulting corrections remain in the auxiliary polydisc and are as small as the original mismatch. Uniform continuity on the prescribed compact sets gives the asserted approximation. $\square$

The domains in this lemma may be thin neighborhoods of compact planar sets. In applications below the cutoff is chosen first on a fixed overlap band. Its derivative bound, and the diameter bound in Lemma 2.4, do not increase when the domains are subsequently narrowed away from that band.

2.3 One planar enlargement with passive parameters

Proposition 2.9. *Let K be a closed circular disc of radius r , or a closed circular annulus with outer radius r , and let f be holomorphic near $K \times P$, where P is a closed polydisc. Suppose that for every ζ on the outer circle there is a coordinate witness neighborhood V_ζ for L containing $f(\{\zeta\} \times P)$. For every $R > r$ and $P_0 \Subset \text{int } P$, one can approximate f uniformly on $K \times P_0$ by a map holomorphic near the enlarged disc, respectively annulus with the same inner radius and outer radius R , times P_0 .*

Proof. We give a finite construction with explicit approximation margins. Choose a compact intermediate parameter polydisc strictly between P_0 and P . Compactness gives a partition of the outer circle into finitely many closed arcs, with endpoint margins, such that all values over each slightly enlarged arc and the intermediate parameter set lie in one coordinate witness neighborhood V_j . At the common endpoint of consecutive arcs the compact set of values lies in $V_j \cap V_{j+1}$ with a positive target margin. Take a slightly larger outer radius than R during the construction.

Attach radial spokes. At each endpoint choose a small filled polar rectangle crossing the old outer circle, on which f and a chart spray from Lemma 2.6 have their values in both adjacent target charts. The rectangles can be chosen mutually disjoint. Attach to each rectangle a radial segment reaching the larger outer radius. In one of its target charts, continue the coordinate spray constantly along the segment from its attachment point on the outer edge of the rectangle. This

defines continuous coordinate functions on the filled rectangle and attached segment, holomorphic on its interior and holomorphic in the passive and auxiliary parameters.

The planar compact set has connected complement. Mergelyan approximation, with parameters, approximates these coordinate functions by polynomials. Here the parameter assertion follows directly by taking finite Taylor sums on a slightly larger parameter polydisc and applying scalar Mergelyan to their coefficients. Consequently the approximation can be made uniform with an auxiliary-parameter margin, and Cauchy estimates preserve the required spray derivatives. On the radial segment its values remain close to the endpoint values for all parameters. A thin neighborhood of each segment therefore also has its values in both adjacent V_j 's.

Glue this extension to the old chart spray across a radial band inside each original rectangle, using Lemma 2.8. More explicitly, cut the union domain at a radius slightly larger than r , still inside the old map domain. At the cut it consists only of the disjoint rectangles. A radial cutoff with a fixed transition width gives separated fringes and a bounded splitting constant. The tubes around the longer portions of the spokes may then be narrowed without changing that constant. Gluing errors can be as small as desired. This produces a map near K and all spokes, close to f on K , with all spoke values still in the two adjacent target charts.

Fill the sectors. List the finitely many closed polar sectors between successive spokes. At a given step let K_{old} be the union of the original compact set, all spokes, and the sectors already filled. The next sector S meets K_{old} exactly on its bottom arc and its two radial sides. Along these three sides all map values, with a parameter margin, belong to the chosen V_j . This is true initially by the spoke construction, and it remains true after each sufficiently accurate gluing, because the sides of every unfilled sector already belong to K_{old} .

Choose a thin closed U-shaped band D about these three sides, inside the current map domain, and a slightly enlarged filled sector D' containing S in its interior, so that $D \subset D'$ is a special pair. To see this elementary geometry, straighten the sector to a rectangle: D is a thick band around its bottom and two side edges, and the missing central rectangle attaches along the U-shaped portion of its boundary. The bands and protrusions can be made as thin as needed. All values near D lie in V_j . Apply L to the restriction of a chart spray over the current map, regarding its small auxiliary variables as additional passive parameters. After any prescribed small parameter loss this gives a spray on D' , arbitrarily close to the old spray on D .

We spell out the overlap used to glue. Fix an open band $W \subseteq \text{int } D$ containing $K_{\text{old}} \cap S$. Using a slightly smaller band first, the compact remainders on opposite sides are positively separated. A smooth cutoff can therefore be zero near $K_{\text{old}} \setminus W$ and one near $S \setminus W$. On a sufficiently thin neighborhood of $K_{\text{old}} \cup S$, its sublevel and superlevel sets give two open domains with separated fringes and overlap contained in W . They may be chosen so that any point of K_{old} in the new side also lies in the approximation band. The old spray is defined on the first domain, the approximating spray on the second, and they are close on their overlap with an auxiliary margin. Lemma 2.8 produces the required new map, close to the old map on all of K_{old} .

There are only finitely many spokes and sectors. Fix in advance a finite nested sequence of parameter polydiscs between the intermediate one and P_0 . At each step choose errors smaller than the remaining compact target-chart margins and allocate a summable finite part of the requested final error. These choices preserve every unfilled sector's joining sides. No chart condition on the outer boundary of a newly filled sector is required. At the end we have a map on a neighborhood of the desired enlarged compact set times P_0 , with the stated approximation. $\square$

Corollary 2.10 (Propagation across a small disc). *Suppose a coordinate neighborhood contains a closed affine complex-line disc centered at x , with its boundary in the open locus where L holds.*

Then L holds at x . In particular, if the complement of that locus is contained locally in a proper analytic subset, L holds everywhere.

Proof. Slightly tilt the line in two different directions through x . Openness and compactness of the original boundary imply that the two tilted discs still have boundary in the good locus, and their tangent directions at x are transverse. Thicken each disc in a complementary coordinate by a sufficiently small fixed parameter disc. At every boundary point the values for all parameters then lie in one witness neighborhood; finitely many boundary patches give one common positive transverse radius.

For each integer n , apply Proposition 2.9 once to each thickened disc, enlarging its long coordinate to radius n , retaining a fixed smaller transverse radius, and making its error on the initial product less than $1/n$. The resulting finite strips converge near their centers to the original coordinate charts. Their limiting long directions are transverse. Remark 2.5 and the finite-length case of Proposition 2.2 imply L at x .

If a proper analytic subset E contains the exceptional points in a coordinate neighborhood of x , choose a complex line through x not contained in E . Its intersection with E is discrete near x , so a sufficiently small circle on that line avoids E . The first assertion applies. $\square$

Remark 2.11. The proof also records a useful uniform version: finitely many coordinate discs whose boundaries lie in fixed compact subsets of the good locus retain the required boundary containment after sufficiently small deformations of the coordinate charts. All arguments in this section concern one planar variable with passive parameters. No approximation theorem for arbitrary multidimensional source compacta has been used.

3 Approximation on polydiscs

We now pass from the planar operation L to approximation in every source dimension. The first step is to enlarge a polydisc. A change of source coordinates will put the part beyond the original map domain into the form of one long planar variable with bounded passive variables. Near the joining annulus, the passive variables describe sets of arbitrarily small diameter, so that L applies to their images in a single witness neighborhood. We then join this extension to the original map on a buffered open overlap.

Write

$$\overline{\Delta}_R^m = \{z \in \mathbb{C}^m : \max_j |z_j| \leq R\}, \quad \|z\|_\infty = \max_j |z_j|.$$

All holomorphic maps on compact sets below are defined on open neighborhoods of those sets. Extra polydiscs always have positive radii. A zero-dimensional polydisc is allowed.

Theorem 3.1. *Let Y be a complex surface with a complete compatible distance. If the local operation L holds at every point of Y , then Y has the convex approximation property in every source dimension.*

This section proves the polydisc case. Section 4 will remove the remaining restriction on the shape of the convex set.

3.1 Gluing on a buffered open overlap

The source change need not preserve product overlaps, so the planar splitting of Lemma 2.4 must be replaced by a splitting on a convex neighborhood in $\mathbb{C}^n$. Let $Q \subset \mathbb{C}^n$ be compact and convex,

and put $Q_r = \{z : \text{dist}(z, Q) < r\}$ for $r > 0$. An auxiliary variable v ranges over a ball $B_b \subset \mathbb{C}^N$; the splitting operators below act only on z and preserve holomorphic dependence on v .

Suppose two open sets A, B cover a neighborhood of $\overline{Q_r}$ and have a fixed buffered overlap: there is a smooth cutoff equal to zero near $A \setminus B$ and one near $B \setminus A$, whose transition lies inside $A \cap B$ with a positive collar. The collar is understood relative to the total domain; after shrinking Q_r its width has a positive lower bound depending only on the fixed geometry. For a bounded holomorphic jump u , the functions χu and $(\chi - 1)u$, extended by zero where their respective multipliers vanish, have the same $\bar{\partial}$ derivative. Fix $0 < h < \min(r/2, 1)$. The constants below depend only on the fixed overlap geometry, a bounded range of r , and the dimensions, not on the jump u or its holomorphic auxiliary parameters. Solve this common equation on an intermediate convex thickening, using the weighted minimal L^2 solution with weight $|z|^2$ [22, Theorem 2.2.1']. The weight is bounded above and below on the fixed bounded domains, so the solution has L^2 norm at most a fixed constant times that of the input form. Cauchy estimates on the buffered transition region control any fixed number of derivatives of that form by powers of $h^{-1}\|u\|_\infty$. Interior elliptic estimates, applied to $\Delta w = 4 \sum_j \partial \eta_j / \partial z_j$ when $\bar{\partial} w = \sum_j \eta_j d\bar{z}_j$, give the corresponding supremum bound for w on the smaller thickening. Subtracting w from the two cutoff functions gives bounded linear splittings on $A \cap Q_{r-h}$ and $B \cap Q_{r-h}$, with bound $Ch^{-p}\|u\|_\infty$ for some fixed integer p . The minimal solution operator is linear and fixed on the base domain; it therefore preserves holomorphic auxiliary parameters.

Lemma 3.2 (Small nonlinear gluing on an open overlap). *In the preceding buffered open-overlap geometry, let two holomorphic sprays into a complex manifold be related on their overlap by a fiber coordinate change*

$$\gamma(z, v) = v + c(z, v), \quad v \in B_b.$$

Suppose $\|c\|_\infty$ can be made arbitrarily small with $b > 0$ fixed. After any prescribed positive loss of base and fiber margins, the two sprays admit fiber reparametrizations close to the identity which make them agree. Evaluating at $v = 0$ gives a glued holomorphic map. It is arbitrarily close to the original map on each compactum where the corresponding input spray is fixed, or has a uniform modulus of continuity in the fiber variable. No such uniform target estimate is asserted on a side whose spray varies without this control.

Proof. Use the additive splitting to find a, b with $b - a = c$. Shrink the base and fiber domains by h . The splitting bounds and Cauchy estimates in v imply, with a fixed integer p enlarged as necessary,

$$\|a\| + \|b\| \leq Ch^{-p}e, \quad e = \|c\|,$$

including their needed fiber derivatives on intermediate smaller balls. For $Ch^{-p}e$ sufficiently small relative to these margins, $\text{id} + b$ has a fiber inverse on a smaller ball, uniformly in the base. The new transition is

$$(\text{id} + b)^{-1} \circ \gamma \circ (\text{id} + a) = \text{id} + d.$$

The equation determining d is

$$d = c(v + a) - c(v) - \{b(v + d) - b(v)\}.$$

The first difference is quadratic in e , up to powers of h^{-1} . The operation in braces is contractive on a small supremum-norm ball, by the first fiber derivative bound. It follows that

$$\|d\| \leq Ch^{-p}e^2. \tag{3.1}$$

These estimates involve a fixed finite number of derivatives. Thus one common exponent p works throughout the iteration; no derivatives of increasing order are requested.

Choose losses $h_i = h_0 2^{-i}$ whose sum fits within the prescribed margins, leaving intermediate buffers at each step. With $A = Ch_0^{-p}$, Equation (3.1) reads $e_{i+1} \leq A 2^{pi} e_i^2$. Put $E_i = A 2^{p(i+1)} e_i$. Then $E_{i+1} \leq E_i^2$. Choosing $E_0 < 1/2$ gives doubly exponential decay. In particular all sums $\sum_i h_i^{-q} e_i$ converge for every fixed q , and are as small as desired with e_0 .

The successive fiber reparametrizations are composed on the nested domains. Their displacements fit within the reserved buffers and their fiber derivatives have summable deviations from the identity. The compositions therefore converge uniformly. Their limiting transitions are the identity. Composing the two original sprays with these limiting side maps gives equal maps on the overlap. The target approximation assertion follows from the stated continuity control and the smallness of the limiting fiber corrections. $\square$

3.2 A change of coordinates with a thin parameter range

Enlarging one coordinate of a polydisc directly leaves an entire transverse polydisc over each boundary point; its image need not fit one witness neighborhood for L . We instead deform the source by a polynomial automorphism. It will be nearly the identity on the old polydisc, while the rest of the enlarged source has coordinates (s, ξ) with s planar and ξ bounded. The following estimate constructs these coordinates on inverse images of the unit polydisc. The factors d^{-1} in their formula will then make the x -image thin as ξ varies with s fixed on the joining annulus.

Lemma 3.3. *Fix $m \geq 2$, numbers*

$$1 < r_m < \cdots < r_2 < r_1, \quad 0 < a < 1, \quad r_j < ar_{j-1} \quad (2 \leq j \leq m).$$

For an integer $d \geq 2$, set

$$P_j(w) = (w/r_j)^d, \quad H_d(x) = (ax_m, P_2(x_2) - ax_1, \dots, P_m(x_m) - ax_{m-1}).$$

There are constants $C \geq 1$, $q \in (0, 1)$, and d_0 , depending only on the displayed fixed data, with the following properties. For $d \geq d_0$, $N \geq 1$, and

$$E_{d,N} = H_d^{-N}(\overline{\Delta}_1^m),$$

one has

$$\max_{j \geq 2} |P_j(x_j)| \leq C \max(1, |x_1|) \quad (x \in E_{d,N}). \quad (3.2)$$

On $E_{d,N} \cap \{|x_1| > r_1\}$, all x_j are nonzero and

$$\left| \frac{H_d(x)_j}{ax_{j-1}} \right| \leq q \quad (2 \leq j \leq m). \quad (3.3)$$

Consequently the holomorphic functions

$$\xi_j(x) = \log \left(\frac{P_j(x_j)}{ax_{j-1}} \right), \quad 2 \leq j \leq m,$$

using the logarithm on $\{|\zeta - 1| < 1\}$, satisfy

$$|\xi_j(x)| \leq B := -\log(1 - q).$$

Here the assertions about holomorphicity are on any open inverse image of the unit polydisc; all the estimates also hold on its closure. With

$$D = d^{m-1}, \quad \rho_1 = 1, \quad \rho_j = r_j(a\rho_{j-1})^{1/d},$$

there is a single-valued holomorphic function s on this end such that

$$x_j = \rho_j s^{d^{m-j}} \exp\left(\sum_{k=2}^j \frac{\xi_k}{d^{j-k+1}}\right), \quad 1 \leq j \leq m. \quad (3.4)$$

For $j = 1$, the sum is empty and this says $x_1 = s^D$.

Proof. Put $r_* = r_1$. Choose $S > \max(2, r_*)$, and then $d_0 \geq 2$ large enough that

$$(M/r_*)^d \geq 4M \quad (M \geq S, d \geq d_0).$$

For example, it suffices to impose this at $M = S, d = d_0$, since $(M/r_*)^d/M$ increases in both variables on this range. The cone

$$\mathcal{E} = \{u : \max_{j \geq 2} |u_j| > \max(S, |u_1|)\}$$

is forward invariant under H_d . Indeed, if $M = \max_{j \geq 2} |u_j|$ and $|u_k| = M$, then

$$|H_d(u)_k| \geq (M/r_*)^d - aM \geq 3M, \quad |H_d(u)_1| \leq aM.$$

The transverse maximum therefore grows by at least a factor 3. In particular, no forward orbit that ends in $\overline{\Delta}_1^m$ can enter $\mathcal{E}$.

Choose C larger than

$$1, \quad 4a, \quad 16a^2 r_*^2, \quad 4S.$$

If $A = \max_{j \geq 2} |P_j(x_j)| > C \max(1, |x_1|)$, then

$$\max_{j \geq 2} |x_j| \leq r_* A^{1/d} \leq r_* \sqrt{A}.$$

For an index k attaining A , both possible terms $a|x_{k-1}|$, namely $a|x_1|$ when $k = 2$ and a transverse coordinate otherwise, are at most $A/4$. Hence

$$|H_d(x)_k| \geq 3A/4, \quad |H_d(x)_1| \leq ar_* \sqrt{A} \leq A/4.$$

Thus $H_d(x) \in \mathcal{E}$, a contradiction. This proves (3.2); the same argument applies at every nonfinal point of the orbit. At its final point the bound is immediate.

Write $y = H_d(x)$. Applying the power bound at y , or using $\|y\|_\infty \leq 1$ if y is the final point, gives

$$|y_j| \leq r_j (C \max(1, a|x_m|))^{1/d} \quad (j \geq 2). \quad (3.5)$$

Let $L_j = \log^+ |x_j|$, $M = L_m$, and $A_1 = \log(r_* \sqrt{C} + a)$. From $P_j(x_j) = y_j + ax_{j-1}$ and (3.5),

$$|P_j(x_j)| \leq (r_* \sqrt{C} + a) \exp(L_{j-1} + M/d).$$

Whether or not $|x_j|$ exceeds 1, this implies

$$L_j \leq \log r_j + A_1/d + L_{j-1}/d + M/d^2. \quad (3.6)$$

Put $A_2 = \log r_* + A_1/2$. Iterating (3.6) to $j = m$ yields

$$M \leq 2A_2 + d^{-(m-1)}L_1 + M \sum_{i=0}^{m-2} d^{-2-i} \leq 2A_2 + d^{-(m-1)}L_1 + \frac{1}{2}M.$$

For $|x_1| \geq r_1$, define

$$t = (|x_1|/r_1)^{1/D} \geq 1, \quad A_3 = 4A_2 + 2 \log r_1.$$

It follows that

$$M \leq A_3 + 2 \log t, \quad |y_j| \leq r_j \exp((\log C + A_3)/d)t^{2/d}.$$

Choose a sufficiently small $\delta > 0$ so that

$$q := \max_{2 \leq j \leq m} \frac{r_j + \delta}{a(r_{j-1} - \delta)} < 1, \quad r_j - \delta > 0.$$

Increase d_0 , if necessary, so that for all $d \geq d_0$

$$r_j \exp((\log C + A_3)/d) \leq r_j + \delta, \quad r_j((1 - q)a(r_{j-1} - \delta))^{1/d} \geq r_j - \delta.$$

We prove successively that

$$|x_j| \geq (r_j - \delta)t^{d^{m-j}} \quad (1 \leq j \leq m).$$

The assertion for $j = 1$ follows from the definition of t . If it holds for $j - 1$, then

$$\frac{|y_j|}{a|x_{j-1}|} \leq \frac{r_j + \delta}{a(r_{j-1} - \delta)} t^{2/d - d^{m-j+1}} \leq q.$$

Therefore

$$|P_j(x_j)| \geq (1 - q)a|x_{j-1}|,$$

and taking the d -th root proves the next assertion. This also proves (3.3). Since $P_j/(ax_{j-1}) = 1 + y_j/(ax_{j-1})$, the indicated logarithm is well defined; its power series gives the bound $|\xi_j| \leq \sum_{\nu \geq 1} q^\nu / \nu = B$.

There is no root choice in the definition of s . Define it by

$$s = \frac{x_m}{\rho_m} \exp \left(- \sum_{k=2}^m \frac{\xi_k}{d^{m-k+1}} \right).$$

The equations

$$x_j^d = r_j^d a x_{j-1} \exp(\xi_j)$$

then prove (3.4) backwards from $j = m$ to $j = 1$, using the definition of the positive real numbers ρ_j . $\square$

Proposition 3.4. *Under the hypotheses of Theorem 3.1, a map holomorphic near a closed polydisc in $\mathbb{C}^m$ can be approximated uniformly there by entire maps $\mathbb{C}^m \rightarrow Y$. More precisely, a map near $\bar{\Delta}_1^m$ can be approximated there by maps holomorphic near $\bar{\Delta}_R^m$, for every finite $R > 1$.*

Proof. First prove the finite-radius assertion. When $m = 1$, the one-dimensional boundary circle is covered by finitely many of the neighborhoods in L , and Proposition 2.9 applies with no passive variables.

Suppose $m \geq 2$. Let f be the initial map and let $\varepsilon > 0$. Choose $b > 1$ with f holomorphic near $\overline{\Delta}_b^m$. Choose the radii r_j , the number a , and $\ell > 0$ so that Lemma 3.3 applies and

$$1 < r_m < \cdots < r_1 < r_1 + 3\ell < b.$$

The derivative of H_d at zero is the d -independent map

$$L(x) = (ax_m, -ax_1, \dots, -ax_{m-1}), \quad \|Lx\|_\infty = a\|x\|_\infty.$$

The map H_d is an automorphism: from $y = H_d(x)$, first recover $x_m = y_1/a$, and then successively recover

$$x_{j-1} = (P_j(x_j) - y_j)/a, \quad j = m, m-1, \dots, 2.$$

Fix $R_+ > R$, and fix $N \geq 1$ so that $a^N R_+ < 1$. Set

$$h_d = H_d^{-N} L^N, \quad \Omega = \Delta_{R_+}^m.$$

Thus $h_d(\overline{\Omega}) \subset E_{d,N}$.

On some fixed neighborhood of $\overline{\Delta}_1^m$,

$$h_d \longrightarrow \text{id} \quad (d \longrightarrow \infty). \quad (3.7)$$

Here is a quantitative justification. Choose $1 < b_0 < b_1 < r_m$, put $\eta = (b_1 - b_0)/3$, and put $\theta = b_1/r_m < 1$. If $\|L^{-1}y\|_\infty \leq b_0 + \eta$ and $\theta^d/a < \eta$, the inverse recursion above gives

$$\|H_d^{-1}(y) - L^{-1}(y)\|_\infty \leq \theta^d/a.$$

Indeed, its first recovered coordinate is exact, and each next one differs from the corresponding coordinate of $L^{-1}y$ by $P_j(x_j)/a$. Inductively all recovered coordinates have modulus at most $b_0 + 2\eta < b_1$, so that $|P_j(x_j)| \leq \theta^d$. Let $C_N = \sum_{i=1}^N a^{-i}$, and take d large enough that $C_N \theta^d/a < \eta$. Successive comparison of $H_d^{-r} L^N(z)$ and $L^{N-r}(z)$, for $z \in \overline{\Delta}_{b_0}^m$, gives the error recursion

$$e_0 = 0, \quad e_{r+1} \leq a^{-1}e_r + \theta^d/a, \quad e_r \leq \theta^d \sum_{i=1}^r a^{-i}.$$

The imposed bound guarantees the hypothesis of the inverse estimate at every step. At $r = N$ it proves $\|h_d - \text{id}\|_{\overline{\Delta}_{b_0}^m} \leq C_N \theta^d$, and hence (3.7).

Choose a number $b' < b$ larger than $r_1 + 2\ell$. Lemma 2.6 gives a chart spray

$$f^\sharp(x, v), \quad f^\sharp(x, 0) = f(x),$$

on a neighborhood of $\overline{\Delta}_{b'}^m \times \overline{\Delta}_\beta^2$ for some $\beta > 0$. Its vertical differential is invertible and its vertical maps are injective after reducing β . The value b' and this spray are fixed before d is increased.

Write $\Phi_d(s, \xi)$ for the right side of (3.4), now allowing ξ to vary independently. Use, for example, the three parameter polydiscs

$$\Xi_i = \{\xi \in \mathbb{C}^{m-1} : |\xi_j| \leq B + i\}, \quad i = 1, 2, 3.$$

Choose fixed numbers

$$r_1 < c_0 < c_1 < c_2 < c_3 < r_1 + \ell.$$

On

$$c_0 \leq |s|^D \leq c_3, \quad \xi \in \Xi_3,$$

one has, uniformly as $d \rightarrow \infty$,

$$|\Phi_{d,j}(s, \xi)| \leq r_j \exp(A_4/d) \quad (j \geq 2), \quad \sup_{\xi, \tilde{\xi} \in \Xi_3} \|\Phi_d(s, \xi) - \Phi_d(s, \tilde{\xi})\|_\infty \leq A_5/d, \quad (3.8)$$

where the following fixed constants suffice:

$$\begin{aligned} R_0 &= 2 \log r_1 + |\log a|, \\ A_4 &= |\log a| + R_0 + \log c_3 + 2(B+3), \\ A_5 &= 4(B+3)r_1 e^{A_4/2}. \end{aligned}$$

Indeed the recurrence for ρ_j gives

$$|\log \rho_j| \leq R_0, \quad |\log \rho_j - \log r_j| \leq (|\log a| + R_0)/d.$$

The power of s contributes at most $\log c_3/d^{j-1} \leq \log c_3/d$. The remaining exponential satisfies

$$\sum_{k=2}^j |\xi_k|/d^{j-k+1} \leq 2(B+3)/d.$$

Moreover

$$\partial_{\xi_k} \Phi_{d,j} = d^{-(j-k+1)} \Phi_{d,j} \quad (k \leq j), \quad \partial_{\xi_k} \Phi_{d,j} = 0 \quad (k > j),$$

Integrating along line segments in the convex polydisc Ξ_3 and using $\sum_{\nu \geq 1} d^{-\nu} \leq 2/d$ proves the diameter bound with the displayed A_5 . In particular Φ_d takes the indicated product into $\Delta_{b'}^m$ for large d .

The compact set $f(\overline{\Delta_{b'}^m})$ has a finite cover by neighborhoods on which L is available. A Lebesgue number for a slightly smaller such cover, uniform continuity of f , and (3.8) show the following. After increasing d and decreasing the fixed spray radius, all the values

$$f^\sharp(\Phi_d(s, \xi), v), \quad \xi \in \Xi_3, \quad v \in \overline{\Delta_\beta^2},$$

belong to one of these neighborhoods for each fixed s on the outer circle $|s|^D = c_3$. The chosen neighborhood may depend on s . The same increase of d ensures that

$$\sup_{\overline{\Delta_1^m}} d_Y(f(h_d(z)), f(z)) < \varepsilon/2, \quad |h_d(z)_1| < c_1 \quad (z \in \overline{\Delta_1^m}).$$

Require also $r_j(Cc_2)^{1/d} < b'$ for $j \geq 2$; this is possible since $r_j < b'$. Fix this value of d , henceforth writing $h = h_d$.

Apply Proposition 2.9 to the annulus

$$A_s = \{c_0 \leq |s|^D \leq c_3\}$$

and to the map $f^\sharp(\Phi_d(s, \xi), v)$, with ξ, v as passive variables. It gives arbitrarily close approximation on $A_s \times \Xi_2 \times \overline{\Delta_{\beta/2}^2}$ by a holomorphic map $G(s, \xi, v)$ defined out to any prescribed finite outer

radius in s , with a smaller fixed positive parameter margin. Choose that radius larger than $c_3^{1/D}$ and larger than

$$\sup\{|s(h(z))| : z \in \overline{\Omega}, |h(z)_1| \geq c_1\}.$$

If the set is empty the latter requirement is vacuous. Otherwise this supremum is finite by compactness and the explicit formula for s ; $|s(h(z))|^D = |h(z)_1|$. The inner radius of the domain of G can be kept strictly below $c_1^{1/D}$.

On the two open subsets of Ω

$$A = \{z : |h(z)_1| < c_2\}, \quad B' = \{z : |h(z)_1| > c_1\},$$

consider the maps with the common auxiliary variable v

$$F_A(z, v) = f^\sharp(h(z), v), \quad F_{B'}(z, v) = G(s(h(z)), \xi(h(z)), v).$$

The first is defined on all of A : by (3.2), its transverse coordinates satisfy

$$|h(z)_j| \leq r_j(Cc_2)^{1/d} < b'$$

after the choices already made. The second is defined on all of B' by Lemma 3.3 and the choices of the parameter and outer-radius margins. On their overlap, $\Phi_d(s(h(z)), \xi(h(z))) = h(z)$; therefore $F_{B'}$ approximates F_A arbitrarily closely, uniformly for v in a fixed smaller polydisc.

Choose a smooth cutoff of $|h(z)_1|^2$ whose transition is compactly contained in $c_1 < |h(z)_1| < c_2$. On the bounded domain Ω , all its derivatives needed in Lemma 3.2 are bounded. These constants, d, N , and all the domain margins have been fixed before choosing the accuracy in Proposition 2.9. Lemma 3.2 therefore glues the two maps, by arbitrarily small vertical corrections, on a neighborhood of $\overline{\Delta}_R^m$. On $\overline{\Delta}_1^m \subset A$ the resulting central map differs from $f \circ h$ by less than $\varepsilon/2$. Together with (3.7), this proves the finite-radius assertion.

To obtain an entire map, use this assertion, after affine rescaling, successively on exhausting polydiscs. Choose positive errors with sum less than the prescribed final error, and ensure that the error at each stage is controlled on the previous polydisc. Completeness makes the maps uniformly Cauchy on every compact set. Their limit is holomorphic: near each point, choose a coordinate neighborhood of the limit value; uniform convergence puts all sufficiently late maps in that chart on a smaller source neighborhood, where the usual theorem for uniformly convergent holomorphic functions applies. The same argument gives the prescribed approximation on the initial polydisc. $\square$

4 From polydiscs to convex approximation

We now have approximation on polydiscs in every dimension. Convex sets need not fit a polydisc inside the original map domain, so this alone does not prove CAP. The required enlargement has the form

$$Q_- = Q \cap \{\operatorname{Im} z_1 \leq 0\} \subset Q,$$

where Q is compact and convex. Given a map near Q_- , we will first approximate it on the joining real face by an entire map, with control of tangential derivatives. That map supplies values on the upper side. Its accuracy need not persist on any fixed complex collar of the face, so the second step is a gluing estimate measured on the face itself. Repeating this enlargement across the faces of a polytope will prove Theorem 3.1.

The induction must retain arbitrary extra polydisc factors: they include both genuine source parameters and the auxiliary variables of the chart sprays used to glue maps.

For $r \geq 0$, denote by $\mathcal{A}_r$ the following statement: for every compact convex $C \subset \mathbb{C}^r$, every extra closed polydisc P , and every map holomorphic near $C \times P$, there are entire maps approximating it uniformly on $C \times P$. All the approximations in $\mathcal{A}_r$ may also be required in any fixed finite number of derivatives on a slightly smaller compact product. To justify this last formulation, first thicken C, P within the given domain, approximate uniformly there, and use chart-spray coordinates and Cauchy estimates on the smaller product. This uses no new approximation assertion.

4.1 Holomorphic approximation with a real parameter

The slices of a convex real face vary with its real coordinate. We will join approximations on neighboring slices by a smooth real-parameter path. The next lemma makes such a path holomorphic near the real interval; the following lemma then makes it entire.

Lemma 4.1 (Complexifying a real parameter). *Let $I \subset \mathbb{R}$ be a compact interval and let $C \subset \mathbb{C}^k$ be a compact convex set. Suppose $H_s(q)$ is a smooth family, for s in a neighborhood of I , of holomorphic maps on a common neighborhood of C , with values in a complex surface. Smoothness is understood on a common neighborhood with all finite derivatives under consideration. For every finite j and every $\epsilon > 0$, after an arbitrarily small loss of the specified compact margins, H can be approximated in $C^j(I \times C)$ by a map holomorphic on a neighborhood of $I \times C$ in $\mathbb{C} \times \mathbb{C}^k$. The size of this complex neighborhood may depend on the approximant. Additional closed polydisc factors can be included in C , with fixed larger neighborhoods.*

Proof. We give the gluing detail to avoid assuming a holomorphic extension of the original smooth family. By Lemma 2.6, a finite cover of I by short intervals admits fixed graph charts $E_i(q, v)$ over the corresponding maps. First choose a smooth family of small chart sprays $H_s^\sharp(q, v)$ centered at $H_s(q)$, holomorphic in (q, v) , on a common small fiber ball. This can be done explicitly along the interval. In one fixed chart, translate its fiber coordinates to the center representing H_s . When changing charts at a division point s_0 , express the preceding spray in the new chart and take its fiber displacement at s_0 . On a short overlap interpolate this displacement to the fixed displacement at s_0 , leaving the moving center unchanged. The two displacements are as close in the first fiber derivative as desired when the overlap is short. The derivatives therefore remain invertible. Finitely many such operations and a smaller common fiber ball give the asserted smooth spray family.

In each division interval write this family in one fixed graph chart. Approximate its coordinate functions by polynomials in s with coefficients holomorphic in (q, v) , retaining any prescribed finite number of derivatives, and interpolate its endpoint jets through order M . This is ordinary approximation in a Banach space of holomorphic functions on a smaller fixed neighborhood: approximate a sufficiently high real derivative uniformly, integrate, and make a finite Hermite correction at the endpoints. The correction tends to zero with the initial approximation error. The target sprays obtained on adjacent intervals consequently have identical s -jets through order M at the division point.

Now, after the finitely many polynomials have been fixed, choose a small complex thickness $\delta > 0$ around the real intervals. Extend each interval slightly past its endpoints, so adjacent rectangles overlap within distance $O(\delta)$ of their common division point. Their spray transition differs from the identity by $O(\delta^{M+1})$ there, uniformly on a fixed smaller fiber ball and parameter compactum. This is Taylor's formula applied to the holomorphic coordinate transitions; the constants are fixed before δ is chosen.

Additive splitting on these finitely many thin planar rectangles costs at most $C\delta^{-1}$. Indeed, use cutoffs across the vertical overlap bands, whose first derivatives are $O(\delta^{-1})$, and the Cauchy–Green operator on a bounded planar set. For simultaneous splitting, each cut contributes a pair of corrections on the entire portion to its left and right; that contribution has zero jump at every other cut. The number of cuts is fixed. The source variables (q, v) are passive for this operator, so no parameter radius is lost.

For clarity, write the consecutive transitions as $\gamma_i = \text{id} + c_i$, with $\|c_i\| + \|D_v c_i\| \leq A\delta^{M+1}$ on a fixed smaller fiber ball. Let R be the simultaneous right inverse, of norm at most K/δ , for the jump operator $(a_i) \mapsto (a_{i+1} - a_i)$. The central maps are identified by the fixed-point equation

$$(a_i) = R((c_i(\cdot, a_i))_i).$$

It preserves the ball of radius $2KA\delta^M$ and is contractive there when $KA\delta^M < 1/2$. The corrections are therefore $O(\delta^M)$. Substituting them into the corresponding sprays gives central maps agreeing on overlaps. Cauchy estimates on rectangles with margins comparable to δ bound their correction derivatives through order j by $O(\delta^{M-j})$. Derivatives in the other variables use their fixed margins. Take $M > j + 2$ and then δ small. Together with the initial polynomial accuracy this gives the prescribed C^j estimate. $\square$

Lemma 4.2. *Assume $\mathcal{A}_r$. Let $I \subset \mathbb{R}$ be a compact interval, $C \subset \mathbb{C}^r$ compact and convex, and P a closed polydisc. A map holomorphic near $I \times C \times P$ can be approximated there, with any prescribed finite derivative accuracy and smaller positive margins, by entire maps of all the displayed complex variables.*

Proof. First shrink any unused margins so that the given map H is defined on $V \times U$, where $V \subset \mathbb{C}$ is a simply connected bounded neighborhood of I , and U contains a slightly thickened compact product $C \times P$. A thin open rectangle is a possible V . Let $\phi : V \rightarrow \Delta$ be a Riemann map. Choose $0 < \rho < 1$ so that $\phi(I) \Subset \Delta_\rho$. The map

$$(\zeta, y, u) \mapsto H(\phi^{-1}(\zeta), y, u)$$

is defined near $\overline{\Delta}_\rho \times C \times P$, with slightly smaller positive margins. Apply $\mathcal{A}_r$, using the ζ disc as an extra passive factor. This gives an entire map $E(\zeta, y, u)$ with arbitrarily good derivative approximation on a smaller product still containing $\phi(I) \times C \times P$. Consequently $E(\phi(s), y, u)$ approximates $H(s, y, u)$ with the required derivatives along $I \times C \times P$.

Choose a closed convex neighborhood I' of I contained in V . Scalar polynomial approximation, with Cauchy estimates after a further harmless shrink, approximates ϕ and any specified finite number of its derivatives on I' by a polynomial p . At this stage E is fixed. On the compact sets in question all its derivatives up to the required order are bounded. The chain rule therefore shows that $E(p(s), y, u)$ is as close to $E(\phi(s), y, u)$ as desired. The map $E(p(s), y, u)$ is entire in every variable. $\square$

4.2 Approximation on a convex real face

Lemma 4.3. *Assume $\mathcal{A}_{n-1}$, where $n \geq 1$, and Proposition 3.4. Let $F \subset \mathbb{R} \times \mathbb{C}^{n-1}$ be compact and convex, and let P be a closed polydisc. Every holomorphic map near $F \times P$ can be approximated on $F \times P$ by entire maps on $\mathbb{C}^n \times \mathbb{C}^{\dim P}$, with any fixed finite derivative accuracy along the real face and smaller positive margins.*

Proof. Write the variables as (t, y, u) , with t real on the face. Thicken F slightly within $\mathbb{R} \times \mathbb{C}^{n-1}$, and thicken P , so that all eventual estimates take place strictly inside the initial domain. Denote the thickened face again by F . Its projection is a compact interval I , and write $F_t = \{y : (t, y) \in F\}$.

We first specify the finite cover used below. For each $t_0 \in I$, choose a compact convex neighborhood C_{t_0} of F_{t_0} in the y variables and an open complex disc D_{t_0} about t_0 such that the given map f is defined near

$$\overline{D}_{t_0} \times C_{t_0} \times P,$$

including some fixed smaller margins. These choices are possible by compactness of $F_{t_0} \times P$. After reducing the real interval about t_0 , it has

$$F_t \Subset \text{int } C_{t_0}$$

uniformly on that interval. Indeed, otherwise a convergent sequence $(t_\nu, y_\nu) \in F$, with $t_\nu \rightarrow t_0$, would contradict $F_{t_0} \subset \text{int } C_{t_0}$. Choose a finite subdivision of I , with small overlaps at its division points, subordinate to these intervals. Denote the corresponding products by $D_i \times C_i \times P$. All sets can be chosen a little larger than the sets used in the subsequent estimates.

At a division point b , both adjacent C_i 's contain F_b in their interiors. Choose compact convex sets C^- and C with nonempty interiors such that

$$F_b \subset \text{int } C^-, \quad C^- \Subset \text{int } C, \quad C \Subset \text{int}(C_i \cap C_{i+1}).$$

By the preceding compactness argument, after reducing a complex disc D_b about b , the sets F_t , for real t near b , lie strictly inside C^- , and

$$\overline{D}_b \times C \times P \Subset (D_i \times \text{int } C_i \times \text{int } P^+) \cap (D_{i+1} \times \text{int } C_{i+1} \times \text{int } P^+)$$

for an available larger passive polydisc P^+ . All transition intervals and their cutoff functions are fixed now. Also fix a large y -polydisc B containing all the C_i and all face projections, with a positive margin.

By $\mathcal{A}_{n-1}$, applied with the t disc and u polydisc passive, choose entire maps $g_i(t, y, u)$ arbitrarily close to f , with any fixed finite derivative accuracy, on $\overline{D}_i \times C_i \times P$. More explicitly, apply the approximation assertion on slightly enlarged closed products still in the given domain and use Cauchy estimates on the displayed products. We join two adjacent g_i 's over one transition, with control on its required face slices.

On a slightly enlarged $D_b \times C \times P$, a chart spray over f , from Lemma 2.6, expresses the two maps as

$$g_i = f^\sharp(\cdot, a_i), \quad g_{i+1} = f^\sharp(\cdot, a_{i+1}),$$

where a_i, a_{i+1} and their required derivatives are as small as desired. Choose a smooth function $\chi : [0, 1] \rightarrow [0, 1]$ equal to 0 near 0 and 1 near 1, and set

$$f_s = f^\sharp(\cdot, (1 - \chi(s))a_i + \chi(s)a_{i+1}).$$

This is a smooth s -family, holomorphic in (t, y, u) . It equals the entire map g_i near $s = 0$ and g_{i+1} near $s = 1$; on these subintervals use those entire definitions on all y . It is arbitrarily close to f on the local product, and its positive s -derivatives up to any fixed order are arbitrarily small there. The constants here depend only on the previously fixed local chart and cutoffs.

The path f_s is defined only over the small convex set C , except near its endpoints, where it equals an entire map. We need a path defined over the common large polydisc B , still close to f_s

on C^- , and equal to the prescribed endpoint maps on all of B . Direct interpolation in the chart over f cannot provide this, because that chart is not defined over B .

We obtain the larger domain by expanding the y variable only where the path already equals an entire endpoint map. After entire approximation on the fixed small set C , we undo that expansion. Choose $d_* \in \text{int } C$. For some $L \geq 1$,

$$d_* + L^{-1}(B - d_*) \subseteq \text{int } C. \quad (4.1)$$

Choose a smooth positive function $\lambda : [0, 1] \rightarrow [1, L]$ which equals 1 outside the endpoint intervals where f_s is an entire endpoint map, and equals L on smaller endpoint intervals. All of these choices are fixed. The family

$$\tilde{f}_s(t, y, u) = f_s(t, d_* + \lambda(s)(y - d_*), u), \quad y \in C,$$

is defined on a common neighborhood of $[0, 1] \times \overline{D}_b \times C \times P$. This assertion uses entirely precisely where $\lambda > 1$; where only the local definition of f_s is available, $\lambda = 1$. Apply Lemma 4.1 to make this family jointly holomorphic near that compact product, with arbitrarily good finite derivative accuracy. Next apply Lemma 4.2, with $\mathcal{A}_{n-1}$ and passive factors D_b, P , to obtain an entire approximant $E(s, t, y, u)$.

For real s , put

$$A_s(t, y, u) = E(s, t, d_* + \lambda(s)^{-1}(y - d_*), u).$$

It is smooth in s and holomorphic in (t, y, u) on the whole prescribed large product. On $C' = C^- \subseteq \text{int } C$, which contains all required overlap slices, the inverse scaling stays in a fixed compact subset of $\text{int } C$: by convexity it lies in the convex hull of $C' \cup \{d_*\}$. The two scalings cancel exactly:

$$\tilde{f}_s(t, d_* + \lambda(s)^{-1}(y - d_*), u) = f_s(t, y, u).$$

Thus approximating $\tilde{f}_s$ on that fixed compact subset makes A_s arbitrarily close to f_s , with the required derivatives, on these slices. On the smaller endpoint intervals, (4.1) gives the stronger statement that A_s is arbitrarily close to g_i , respectively g_{i+1} , on the entire large product $\overline{D}_b \times B \times P$. The chain rule costs only finite constants, since λ and all scaling factors have already been fixed.

We may arrange exact endpoint joins. A chart spray over g_i exists on a slightly larger version of that large product. On a smaller initial s -interval, write $A_s = g_i^\sharp(\cdot, \beta_s)$. Here β_s , including all required derivatives, is as small as desired. Replace β_s by $\kappa(s)\beta_s$, where κ is 0 near 0 and 1 before leaving this endpoint interval. Do the analogous replacement near 1. The repaired path, denoted H_s , is exactly g_i near one endpoint and exactly g_{i+1} near the other, and retains the desired approximation on the overlap slices. It need only be holomorphic on a neighborhood of the fixed large product, which the chart construction provides.

There is no circular smallness requirement in this procedure. First choose the local g_i 's sufficiently accurate that the interpolated f_s has the desired small error after the fixed real transition cutoff. Then, with g_i, λ, B and the endpoint chart sprays fixed, choose the complexification and entire approximation of $\tilde{f}_s$ as accurate as needed to overcome all their finite derivative and chart constants. Finally perform the exact endpoint repairs.

Substitute a fixed smooth cutoff $s = s(t)$ across the real transition interval. The result is a smooth real t -family, holomorphic in (y, u) on the large product, which joins the two local entire maps and is as close to f as prescribed, with all required tangential derivatives, on the face slices. Doing this at the finitely many disjoint transition intervals gives one such family on all of I . Extend it smoothly past the endpoints of I using the first and last entire maps g_i , to which it already agrees on endpoint intervals.

Apply Lemma 4.1 once more, now to this real t -family on the fixed large polydisc in (y, u) . The result is jointly holomorphic near $I \times B \times P$ and retains the prescribed accuracy on the face. Proposition 3.4 is $\mathcal{A}_0$; thus Lemma 4.2 with $r = 0$ converts this last family into an entire map in (t, y, u) , still with the required accuracy. When $n = 1$, there are no y variables and the scaling step is omitted; the same argument, or just this last complexification and $\mathcal{A}_0$ step, applies. $\square$

4.3 Gluing across a real interface

Lemma 4.3 provides an entire upper-side map close to the old map on the real face. Unlike the buffered overlap in Lemma 3.2, the complex neighborhood on which this accuracy persists may depend on the approximant. We therefore solve the additive jump problem using only its Hölder trace. The nonlinear correction will use the same quadratic iteration as open-overlap gluing, now in this trace norm.

Fix $0 < \alpha < 1$. For a compact convex set $Q \subset \mathbb{C}^n$, write

$$Q_r = \{z : \text{dist}(z, Q) < r\}, \quad M_r = Q_r \cap \{\text{Im } z_1 = 0\}, \quad Q_r^\pm = Q_r \cap \{\pm \text{Im } z_1 > 0\}.$$

We always start with $r > 0$, including when Q has empty interior. An auxiliary variable v ranges over a ball $B_b \subset \mathbb{C}^N$. All functions under consideration are holomorphic in v . All norms below are uniform in v ; the Hölder seminorm is taken in the base variables only. Traces on M_r are holomorphic in $z' = (z_2, \dots, z_n)$ on every available slice. For a side function set

$$\|a\|_{r,b,*} = \sup_{Q_r^\pm \times B_b} |a| + \sup_{v \in B_b} \|a(\cdot, v)\|_{M_r} \|C^\alpha.$$

The Hölder norm on an open bounded set means the supremum norm plus the supremum of all difference quotients in that set. The constants in this section may depend on a bounded range of r , the dimensions, α , and the fixed joining geometry, but not on the input functions.

Lemma 4.4 (Additive splitting at a real interface). *There are $C > 0$ and an integer $p \geq 1$ with the following property. For $0 < h < \min(r/2, 1)$ and a vector-valued function u on $M_r \times B_b$ with the regularity just specified, there are linear operators producing functions $a^\pm$, holomorphic on $Q_{r-h}^\pm \times B_b$ and continuous up to $M_{r-h} \times B_b$, such that*

$$a^+ - a^- = u \quad \text{on } M_{r-h} \times B_b, \quad \|a^+\|_{r-h,b,*} + \|a^-\|_{r-h,b,*} \leq Ch^{-p} \sup_{v \in B_b} \|u(\cdot, v)\|_{C^\alpha(M_r)}.$$

The operators preserve holomorphic dependence on all auxiliary variables.

Proof. Use several intermediate thickenings between Q_r and Q_{r-h} , with successive distances fixed positive multiples of h . Cover the required part of M_r by coordinate boxes of side comparable to h . The enlarged boxes used below are still inside Q_r . One can choose a grid cover with bounded overlap, at most Ch^{-2n} boxes, and a smooth subordinate partition whose derivatives of order j are bounded by $C_j h^{-j}$. Boxes outside a sufficiently large fixed ball are unnecessary.

In one enlarged box, with center (t_0, z'_0) on the interface, take a smooth real cutoff χ supported in its t -interval and equal to one on a smaller interval. For $\text{Im } \zeta \neq 0$ put

$$C_u(\zeta, z', v) = \frac{1}{2\pi i} \int_{\mathbb{R}} \frac{\chi(t) u(t, z', v)}{t - \zeta} dt.$$

The one-variable Cauchy boundary formula gives the jump u on the smaller interval, after assigning the signs of the two boundary values in the usual way. Its Hölder bound follows by subtracting

$u(t_*, z', v)$ in the singular integral near a boundary point t_* : the remaining numerator is bounded by a constant times $|t - t_*|^\alpha$. The same subtraction in the difference of two boundary values gives the C^α estimate. Rescaling the interval to unit length gives a bound with a fixed power of h^{-1} . The integral is holomorphic in (z', v) by dominated integration. Cauchy estimates on slightly smaller transverse boxes control transverse derivatives, with a further fixed power of h^{-1} . Thus the local side functions have the asserted joint Hölder trace bounds.

Denote these local pairs by $c_i^\pm$. Multiply by a smooth partition θ_i in a band around the interface, equal in sum to one on a smaller band, and extend the partition terms into the two sides with support in their boxes. This gives smooth side functions $f^\pm = \sum_i \theta_i c_i^\pm$ with trace jump u . Their ordinary side derivatives $\bar{\partial} f^\pm$ fit together to a single smooth form η across the interface. Here is the point that ensures smoothness despite the original Hölder data. In a smaller box choose one reference pair $c_0^\pm$. Every difference $c_i^\pm - c_0^\pm$ has equal traces, hence extends holomorphically across the cut by continuity and Morera's theorem. Where $\sum_i \theta_i = 1$,

$$\bar{\partial} f^\pm = \sum_i (\bar{\partial} \theta_i) (c_i^\pm - c_0^\pm).$$

The right side is the restriction of a common smooth form. Outside this smaller band, the terms are evaluated a distance comparable to h from their Cauchy singularities. Consequently any fixed number of derivatives of η is bounded by $Ch^{-p_j} \|u\|_{C^\alpha}$. It is closed, since it is locally a $\bar{\partial}$ derivative on the open sides and smooth across their boundary.

Solve $\bar{\partial} w = \eta$ on an intermediate convex domain. A weighted minimal L^2 solution with the fixed strictly plurisubharmonic weight $|z|^2$ has L^2 norm at most $C\|\eta\|_{L^2}$ by the standard L^2 theorem; the weight is bounded above and below on the domains in question; this is the weighted estimate of [22, Theorem 2.2.1'], also treated in [23, 14]. Interior estimates give both a supremum and a first-derivative bound on the next smaller thickening, with polynomial losses in h^{-1} . Indeed, writing $\eta = \sum_{j=1}^n \eta_j d\bar{z}_j$, the distributional equation gives $\Delta w = 4 \sum_j \partial \eta_j / \partial z_j$ componentwise. Interior elliptic estimates on nested balls of radius comparable to h , followed by Sobolev embedding, bound w in C^1 by its L^2 norm and a fixed finite number of derivatives of η , with powers of h^{-1} . This proves the required interior bound and hence the Hölder trace bound for w .

The operator just used is fixed and linear on the base domain. Applied to a holomorphic family of L^2 coefficients, it gives a holomorphic family of L^2 solutions. Interior estimates turn this into ordinary joint holomorphic dependence on v . Now set $a^\pm = f^\pm - w$. Their side $\bar{\partial}$ derivatives vanish, their trace jump is unchanged, and the stated estimate follows by enlarging p to dominate the finitely many derivative and covering losses. $\square$

Lemma 4.5 (Small nonlinear gluing on a real interface). *The conclusion of Lemma 3.2 holds for sprays on Q_r^- and Q_r^+ when their transition is given on $M_r \times B_b$ and is arbitrarily small in*

$$\|c\|_{r,b,\alpha} = \sup_{v \in B_b} \|c(\cdot, v)\|_{C^\alpha(M_r)}.$$

Here the sprays are holomorphic on the open sides, continuous up to the interface, and the transition is holomorphic in (z', v) there. The resulting glued map is holomorphic across the interface. No lower bound is assumed for a complex normal thickness on which the input transition is small.

Proof. Apply Lemma 4.4 to split the transition error as $b - a = c$. With losses h in the base and fiber margins, its bound is $Ch^{-p}e$, where $e = \|c\|_{r,b,\alpha}$; here and below the side corrections are measured by their supremum and trace norms. Cauchy estimates in the fiber variable give the

same bound, with additional fixed powers of h^{-1} , for their first and second fiber derivatives on smaller balls.

As in Lemma 3.2, the corrected transition $(\text{id} + b)^{-1} \circ \gamma \circ (\text{id} + a) = \text{id} + d$ is determined on the interface by

$$d = c(v + a) - c(v) - \{b(v + d) - b(v)\}.$$

The C^α multiplication inequality and integration of the first fiber derivative along line segments bound the first difference by $Ch^{-p}e^2$. To compare its values at two base points, the variation of the derivative at the shifted fiber argument is controlled by the second fiber derivative. The same estimates make the operation in braces contractive in the trace norm. Thus, after increasing the fixed exponent p ,

$$\|d\|_{r-h, b-h, \alpha} \leq Ch^{-p}e^2.$$

Only these finitely many derivatives are required at every step. The losses $h_i = h_0 2^{-i}$ and the rescaled errors E_i in the proof of Lemma 3.2 consequently give the same doubly exponential convergence. The sums of side corrections and their fiber derivatives converge on both sides, including their C^α traces. Their limiting transitions are the identity.

Composing the original sprays with the limiting side maps gives equal continuous traces. Near each interface point, continuity places both images in a common target chart; Morera's theorem across the real hyperplane then proves holomorphicity of the joined map. The old-side approximation follows from the small fiber corrections and the fixed spray's uniform continuity on compacta, exactly as in Lemma 3.2. $\square$

4.4 Removal of convex faces

The face lemma supplies an entire upper-side map with arbitrarily accurate tangential derivatives along the join. It remains to attach that map to the given lower-side map, retaining approximation on the whole old convex set. Repeating this single-face step will reach a polydisc, where Proposition 3.4 applies.

Proof of Theorem 3.1. We prove $\mathcal{A}_n$ by induction. The case $n = 0$ is Proposition 3.4. Suppose $\mathcal{A}_{n-1}$ is known.

We first remove a single real affine constraint. After an invertible complex affine change of the n variables, write the enlarged convex compactum as $Q \subset \mathbb{C}^n$, and the old compactum as

$$Q_- = Q \cap \{\text{Im } z_1 \leq 0\}.$$

Assume a map f is given near $Q_- \times P$. Cuts with no enlargement are trivial. In the nontrivial applications below, Q_- has nonempty interior and the cut meets the interior of Q . Figure 2 shows the two sides to be joined.

Choose small convex thickenings of Q and of P . Their lower sides, together with additional small neighborhoods, can be kept in the domain of f . To verify this point without a transversality assumption on the boundary of Q , note that as $\eta \downarrow 0$,

$$(Q + \eta \overline{B}) \cap \{\text{Im } z_1 \leq 0\} \longrightarrow Q_-$$

in the following sufficient sense: every sequence of points in the left-hand sets has all its limit points in Q_- . Compactness then places these sets in any prescribed neighborhood of Q_- , for small η . Here $\overline{B}$ is the Euclidean closed unit ball.

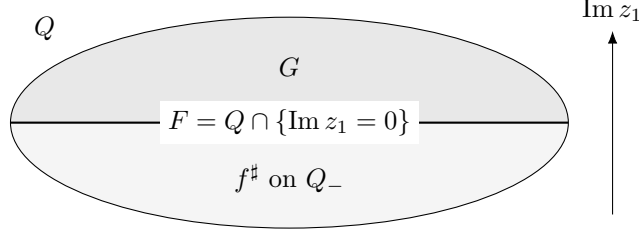


Figure 2: A schematic real section of the single-face enlargement; passive variables are suppressed. The lower region carries $f^\sharp$ and the upper region carries G . The input maps are compared only on a slightly enlarged real face F , with no prescribed complex collar and no bound for G away from the interface.

Lemma 2.6 gives a chart spray $f^\sharp(z, u, v)$ over a neighborhood of the thickened lower side, with a fixed auxiliary polydisc in v . Its trace on

$$\{\operatorname{Im} z_1 = 0\}$$

is holomorphic near a convex compact face in $(t, y) = (\operatorname{Re} z_1, z_2, \dots, z_n)$. Apply Lemma 4.3, treating u, v as passive polydisc variables. We obtain an entire map $G(z, u, v)$ arbitrarily close to $f^\sharp$, with, say, two tangential derivatives on a slightly enlarged face and with fixed smaller u, v margins.

On this interface, invert the vertical chart of $f^\sharp$. For a fixed smaller v -polydisc this produces a transition

$$\gamma(z, u, v) = v + c(z, u, v)$$

between G and $f^\sharp$, with c as small as desired in the C^α trace norm, for any fixed $0 < \alpha < 1$. This follows from the just obtained tangential derivative accuracy and the bounded derivatives of the fixed chart inverse on compact sets; Cauchy estimates on the retained u -margin also give the required Hölder control in u . The transition is holomorphic in the complex tangential variables and in v . Use $f^\sharp$ on the lower side and G on the upper side. Apply Lemma 4.5 with convex base $Q \times P$, normal coordinate z_1 , and only v as the fiber variable. With losses smaller than the chosen thickening margins, it glues their suitably reparametrized central maps on a neighborhood of $Q \times P$. Its corrections can be made arbitrarily small on the lower side, so the resulting map approximates f on $Q_- \times P$. No estimate for G away from the interface is used.

Finally let $K \subset \mathbb{C}^n$ be any nonempty compact convex set, and let f be given near $K \times P$. Thicken K and P slightly within this domain, and choose a full-dimensional real convex polytope K' containing K in its interior and still within the available thickening. Choose a large coordinate polydisc $\overline{\Delta}_R^n$ containing K' in its interior. Write

$$K' = \bigcap_{j=1}^M \{\ell_j(z) \leq b_j\},$$

where the ℓ_j 's are real linear forms, and remove these constraints one at a time inside $\overline{\Delta}_R^n$. At each stage both compact sets are convex, the old one has nonempty interior, and a nonredundant constraint can be put in the form $\operatorname{Im} z_1 \leq 0$ by a complex affine coordinate change. The single-constraint argument therefore extends the map approximately to the next compactum. There are only finitely many stages, so assign their errors a sum smaller than half the desired tolerance. The

final map is defined near $\overline{\Delta}_R^n \times P$, a polydisc. Proposition 3.4 approximates it by an entire map, using the remaining half of the tolerance. This proves $\mathcal{A}_n$, completes the induction, and gives Theorem 3.1 by taking no passive variables. $\square$

5 The projective case

We construct two complete directions from families of genus-one curves. The curves on the K3 surface need not be smooth: the complete flows take place on their smooth normalizations.

Proposition 5.1. *Let S be a smooth projective complex K3 surface. There is a nonempty Zariski open subset $U \subset S$ such that, for every $x \in U$, there are $b > 0$ and holomorphic maps*

$$\sigma_1 : \mathbb{C} \times \Delta_b \longrightarrow S, \quad \sigma_2 : \Delta_b \times \mathbb{C} \longrightarrow S$$

which are locally biholomorphic at the origin and satisfy

$$\sigma_1(0, 0) = \sigma_2(0, 0) = x, \quad \sigma_1(z, 0) = \sigma_2(z, 0) \quad (|z| < b).$$

The complement $S \setminus U$ is algebraic of dimension at most one. Consequently S has property L everywhere.

We use the genus-one case of the theorem of Chen and Gounelas [9, Theorem A]: there are integral curves $C_n \subset S$ of geometric genus one with $C_n^2 \rightarrow \infty$, whose normalization maps belong to smooth proper families of maps

$$p_n : Y_n \longrightarrow T_n, \quad q_n : Y_n \longrightarrow S.$$

Here T_n is an irreducible algebraic curve, the fibers of p_n are connected genus-one curves, and the induced moduli map is nonconstant. One fiber map is the normalization of C_n followed by its inclusion into S . A stable-map family includes, by definition, a proper family of source curves; pulling back that family and restricting to the smooth-source locus gives the proper maps p_n . No properness of T_n is asserted. We use the families over their full proper fibers, even when evaluation is inverted only on a smaller open subset.

Lemma 5.2. *Let S be a smooth projective integral surface, let H be an ample divisor, and let $p : Y \rightarrow T$ be a smooth proper family of connected genus-one curves over a smooth irreducible algebraic curve. Suppose that $q : Y \rightarrow S$ has nonconstant source-curve moduli and that q_{t_0} is the normalization map of an integral curve $C \subset S$. After removing finitely many points of T , every q_t normalizes an integral image curve C_t with $H \cdot C_t = H \cdot C$. The evaluation map q is dominant and generically finite, and some nonempty Zariski open $V \subset S$ has finite etale surjective inverse image map $q^{-1}(V) \rightarrow V$.*

Proof. The total space Y is a smooth integral surface. Smoothness follows from that of p and T ; connectedness follows from connectedness of the base and fibers, and the irreducible components of a smooth variety are disjoint open and closed sets. Proper smooth local triviality shows that $\deg(q_t^*H)$ is constant: it is the integral of $c_1(q^*H)$ on a fiber. At t_0 it equals $H \cdot C > 0$. Thus all fiber maps are nonconstant and finite onto their reduced integral images.

Consider the graph map

$$Q = (p, q) : Y \longrightarrow T \times S.$$

It is proper, by factoring through the closed graph in $Y \times S$ followed by the proper map $p \times \text{id}_S$, and it has finite fibers. The proper quasi-finite criterion therefore makes it finite. Its reduced image Z is an integral surface. Choose a smooth point z_0 of C and its unique preimage $y_0 \in Y_{t_0}$. The fiber $Q^{-1}(t_0, z_0)$ is a singleton. Moreover dQ_{y_0} is injective: a vector in its kernel is vertical for p , and the normalization is an isomorphism near z_0 , so dq is injective on that vertical tangent line.

The holomorphic constant-rank theorem embeds a small neighborhood W of y_0 in $T \times S$. Properness makes $Q(Y \setminus W)$ closed in the analytic topology. Shrink a neighborhood of (t_0, z_0) to avoid that closed set and remain inside the embedding chart. Thus Q is an isomorphism over a nonempty analytic open subset of Z . That subset meets the dense Zariski open locus where the finite map is etale of its generic degree. The degree is therefore one.

Since Y is normal, Q is the normalization of Z . Its nonisomorphism locus $N \subset Z$ is closed of dimension at most one. Remove the finitely many parameters of the vertical curve components of N . Every horizontal component of N has finite intersection with every remaining fiber, so the irreducible support $|Z_t| = q_t(Y_t)$ is not contained in N . The restriction q_t is consequently finite and birational onto its reduced image C_t , hence is its normalization. The projection formula gives

$$H \cdot C_t = \deg(q_t^* H) = H \cdot C.$$

If the closure of $q(Y)$ were a curve, each retained fiber map would normalize that same integral curve. Their smooth source curves would all be isomorphic, contrary to nonconstant moduli. Thus q dominates S . Source and target have dimension two, so the induced function-field extension is finite and, in characteristic zero, separable.

Here the desired finite etale locus must be justified even though q itself is not assumed proper. Choose affine opens $V_0 = \text{Spec } B \subset Y$ and $U_0 = \text{Spec } A \subset S$ with $q(V_0) \subset U_0$. Every one of finitely many A -algebra generators of B is algebraic over $\text{Frac } A$. Invert a common nonzero element of A to make their monic equations integral; the resulting affine map is finite. Remove also the closure of $q(Y \setminus V_0)$, of dimension at most one, and shrink to the etale locus. This leaves a nonempty Zariski open V on which the full inverse image is the finite etale piece. It is surjective because it is finite and dominant. $\square$

Normalize the parameter curves supplied by Chen–Gounelas, retaining a point above each specified normalization fiber, and pull back the families. The bases are then smooth irreducible curves; smoothness, properness, and nonconstant moduli persist. Fix a very ample divisor H on S . The Hodge index inequality

$$(H \cdot C_n)^2 \geq H^2 C_n^2$$

shows that the ample degrees are unbounded. Choose two distinct degrees d_1, d_2 and apply Lemma 5.2. We obtain two families

$$p_i : Y_i \longrightarrow T_i, \quad q_i : Y_i \longrightarrow S \quad (i = 1, 2)$$

whose fiber maps normalize integral curves of degree d_i . Choose a common nonempty open $V \subset S$ on which $Y_i^V = q_i^{-1}(V) \rightarrow V$ are both finite etale and surjective. Only their inverse sheets are restricted to V : the maps q_i remain defined on all of Y_i .

Lemma 5.3. *There is a nonempty Zariski open $U \subset V$ such that any two lifts $y_i \in Y_i^V$ of a point $x \in U$ have distinct transported vertical tangent lines*

$$dq_1(\ker(dp_1)_{y_1}) \neq dq_2(\ker(dp_2)_{y_2}) \quad \text{in } T_x S.$$

The closure in S of the exceptional locus has dimension at most one.

Proof. Form the finite etale cover

$$W = Y_1^V \times_V Y_2^V \xrightarrow{Q} V, \quad \tau_i = p_i \circ \text{pr}_i.$$

Every irreducible component of the smooth surface W dominates V , since a finite etale map is both open and closed. The maps τ_i are submersions. Their algebraic rank-drop locus

$$I = \{d\tau_1 \wedge d\tau_2 = 0\}$$

records equality of the transported vertical directions.

No component of W can be contained in I . If one were, identify a small analytic neighborhood in it with an open subset of S using Q . In a coordinate $s = \tau_1$, dependence of the differentials makes τ_2 depend only on s . A local τ_1 -fiber would then lie in a local τ_2 -fiber. Its nonconstant image germ in S lies in an integral curve from each family. Two distinct integral curves on a projective surface have finite intersection, so the curves must coincide. This contradicts their distinct H -degrees.

Consequently $\dim I \leq 1$. Since Q is finite, $Q(I)$ is closed in V and has dimension at most one; its algebraic closure in S has the same dimension bound. Taking

$$U = S \setminus ((S \setminus V) \cup \overline{Q(I)}^S)$$

proves all assertions. $\square$

Lemma 5.4 (Complete vertical flows). *Let $p : Y \rightarrow \Delta$ be a proper holomorphic submersion with connected genus-one fibers. After shrinking the base disc, there is a nowhere-zero holomorphic vertical vector field whose flow is jointly holomorphic for all complex times. Its restriction to the central fiber may be prescribed as any nonzero holomorphic vector field there.*

Proof. For every fiber C , the space $H^0(C, T_C)$ is one-dimensional and its nonzero elements have no zeros. The relative tangent bundle is locally free and flat over the base. Grauert's cohomology and base-change theorem [19, § 7, Satz 5] makes $p_* T_{Y/\Delta}$ a holomorphic line bundle with these spaces as fibers. The evaluation morphism

$$p^*(p_* T_{Y/\Delta}) \longrightarrow T_{Y/\Delta}$$

is an isomorphism on every fiber, hence everywhere. A local frame gives the vertical field, and scaling it prescribes its central restriction.

Take a smaller closed base disc. Its full inverse image is compact. Local holomorphic flow existence and a finite cover of this compact set give a common time radius $\epsilon > 0$. The flow preserves fibers, so successive small-time flows remain in the same compact inverse image. Composing the time- z/M flow M times defines a holomorphic flow for $|z| < M\epsilon$. Uniqueness and the local group law make these extensions agree as M increases. They define the jointly holomorphic flow for every $z \in \mathbb{C}$ over the smaller open base disc. $\square$

Proof of Proposition 5.1. Fix $x \in U$ and lift y_i as in Lemma 5.3. Use Lemma 5.4 on each full proper family over a small base disc about $p_i(y_i)$. Let ξ_i be the resulting vertical fields and Fl_i^z their complete flows. Then $v_i = dq_i(\xi_i(y_i))$ are independent tangent vectors at x .

Choose a local section $e_1(w)$ of p_1 through y_1 and set

$$\sigma_1(z, w) = q_1(\text{Fl}_1^z(e_1(w))).$$

This map is defined for all $z \in \mathbb{C}$ and all sufficiently small w . It is locally biholomorphic at the origin: the flow derivative is vertical, the section derivative projects nontrivially to the base, and q_1 is etale at y_1 . Put $\varphi(z) = \sigma_1(z, 0)$.

For small z lift $\varphi(z)$ through the local inverse of q_2 at y_2 , obtaining a holomorphic map $\ell_2(z)$ with $q_2(\ell_2(z)) = \varphi(z)$. Shrink its domain so $p_2(\ell_2(z))$ lies in the disc supporting the second complete flow, and define

$$\sigma_2(z, w) = q_2(\text{Fl}_2^w(\ell_2(z))).$$

It is defined for small z and every $w \in \mathbb{C}$. Its derivatives at the origin are v_1, v_2 , so it too is locally biholomorphic. At time zero it returns its starting point, giving the exact identity $\sigma_2(z, 0) = \varphi(z) = \sigma_1(z, 0)$. Choose a common radius for the bounded variables. The inverse sheet was used only for the initial lift; subsequent flow values lie in entire proper fibers, where q_2 is still defined.

Proposition 2.2 gives L throughout U . Its algebraic complement has dimension at most one, so Corollary 2.10 gives L at the remaining points. $\square$

6 Exact symplectic sewing of annular chains

This section isolates the analytic construction needed in the geometric arguments. The domains are cylinders, not discs. Consequently, a closed holomorphic one-form on an overlap need not be exact. Its period is the obstruction that must be removed before the iteration can converge.

Put $\mathcal{C} = \mathbb{C}/\mathbb{Z}$, and write $Y = \text{Im } w$ on $\mathcal{C}$. We use the form

$$\omega_0 = dw \wedge dp$$

on products of $\mathcal{C}$ with a parameter disc. A symplectic map $F = (W, P)$ between such products is called *exact* if

$$F^*(P dW) - p dw$$

is an exact holomorphic one-form. Since a cylinder times a disc has first homology $\mathbb{Z}$, this is equivalent to the vanishing of its integral around one positively oriented period circle. All lifts below have degree one: $W(w + 1, p) = W(w, p) + 1$ and $P(w + 1, p) = P(w, p)$.

Theorem 6.1 (Annular-chain sewing). *Let M be a complex surface with a nowhere-zero holomorphic two-form ω . Fix $0 < h < 1/8$, $r > 0$, and $A < \infty$. For each $j \in \mathbb{Z}$, let $L_j \geq 1$ and let*

$$\phi_j : \{-2h < Y < L_j + 2h\} \times \Delta_r \longrightarrow M$$

be a holomorphic map with $\phi_j^ \omega = \omega_0$. Suppose there are holomorphic translations*

$$J_j(w, p) = (w + a_j(p), p), \quad |\text{Im } a_j(p) + L_j| < h/8, \quad |a'_j(p)| \leq A,$$

and exact symplectic transitions F_j on

$$S_j(h, r) = \{|Y - L_j| < h\} \times \Delta_r$$

such that $\phi_j = \phi_{j+1} \circ F_j$. Write

$$J_j^{-1} F_j = (w + u_j, p + v_j).$$

Assume these lifts exist and $\sup_j \|(u_j, v_j)\|_{S_j(h, r)} \leq e$.

There are $b > 0$ and $e_* > 0$, depending only on h, r, A and the indicated margins, such that, if $e < e_*$, these charts give a holomorphic symplectic immersion

$$\Phi : \mathbb{C} \times \Delta_b \longrightarrow M$$

which is periodic of period one in the first variable. On a smaller domain in every original block, with fixed positive margins at both ends, Φ is obtained from ϕ_j by a small periodic symplectic change of variables. The corrected transitions are

$$(w, p) \longmapsto (w + \tilde{a}_j(p), p).$$

The changes of variables and $\tilde{a}_j - a_j$, together with any fixed finite number of derivatives on further smaller domains, tend uniformly to zero as $e \rightarrow 0$. None of these assertions requires an upper bound for the L_j or a bound on the number of blocks.

Here and below $\Delta_r = \{p \in \mathbb{C} : |p| < r\}$; a norm on a noncompact cylinder is the supremum norm of a periodic function. We give the estimates and the domain construction explicitly.

6.1 Domains and the infinitesimal obstruction

For a current collection of translations, define

$$\begin{aligned} S_j(s, \rho) &= \{|Y - L_j| < s, |p| < \rho\}, \\ D_j(s, \rho) &= \{L_{j-1} - s < Y - \operatorname{Im} a_{j-1}(p), \quad Y < L_j + s, \quad |p| < \rho\}. \end{aligned}$$

The lower edge of D_j is near height zero. The particular slanted lower edge is useful: the lower inequality for D_{j+1} , after composition with J_j , is precisely $Y > L_j - s$. Thus S_j lies in D_j , and its translate lies in D_{j+1} . The other two inequalities have a fixed buffer because $L_j \geq 1$ and $s < 1/8$.

If s and ρ are each decreased by δ , then the smaller D_j admits coordinate Cauchy discs of radius at least $c\delta/(1+A)$ in the larger D_j . The same assertion holds for seams and for their translated copies. This follows directly from

$$|a_j(p') - a_j(p)| \leq A|p' - p|.$$

In particular, all fixed-order Cauchy bounds cost only a fixed power of δ^{-1} , independently of j and L_j . In the estimates below constants may depend on h, r, A and on fixed fractions of the initial margins; we take $0 < \delta < 1$.

Lemma 6.2 (Removal of the transverse zero mode). *Suppose $E = (w + u, p + v)$ is exact symplectic on a seam and $\|(u, v)\| \leq e$. After a margin loss δ there are a holomorphic function $u_0(p)$ and a periodic mean-zero function H such that*

$$(u, v) = (u_0, 0) + X_H + R, \quad X_H = (H_p, -H_w), \quad \|H\| \leq Ce, \quad \|R\| \leq C\delta^{-2}e^2.$$

Here u_0 is the zero Fourier coefficient of u .

Proof. Symplecticity and exactness respectively give

$$u_w + v_p = -u_w v_p + u_p v_w, \tag{6.1}$$

$$0 = \int_0^1 v(w, p)(1 + u_w(w, p)) dw. \tag{6.2}$$

The second identity follows by expanding $(p+v)d(w+u) - p dw$ on a horizontal period circle; the integral of pu_w is zero. Consequently its mean $v_0(p)$ satisfies

$$|v_0(p)| \leq C\delta^{-1}e^2.$$

Take the unique mean-zero periodic primitive satisfying

$$H_w = -v + v_0.$$

Integration along one horizontal period, followed by subtraction of the mean, bounds H by Ce . This primitive is holomorphic in p . Equation (6.1) gives

$$\partial_w(u - H_p) = -u_w v_p + u_p v_w - v'_0(p).$$

The right side has zero mean and is $O(\delta^{-2}e^2)$ after one additional fixed fraction of the margin loss. Integrating its mean-zero primitive bounds $(u - H_p) - u_0$ by the same quantity. The other component of R is v_0 . Rescaling the allocation of δ among these finitely many steps proves the stated estimate. $\square$

The role of exactness is now visible. Without Equation (6.2), the first-order constant term in v need not vanish. It cannot be absorbed into a translation that preserves the common transverse coordinate.

6.2 A uniformly bounded Fourier splitting

Lemma 6.3 (Splitting along the whole chain). *Suppose the periodic functions H_j have mean zero, are holomorphic on $S_j(s, \rho)$, and have norm at most e . Assume*

$$|\operatorname{Im} a_j + L_j| \leq \kappa < 1/4.$$

There are periodic holomorphic functions G_j on $D_j(s - \delta, \rho)$ satisfying

$$G_j - G_{j+1} \circ J_j = -H_j. \tag{6.3}$$

on the smaller seams. Their bounds, on domains with the appropriate additional losses, are

$$\|G_j\| \leq C\delta^{-1}e, \quad \|X_{G_j}\| \leq C\delta^{-2}e, \quad \|DX_{G_j}\| \leq C\delta^{-3}e.$$

All constants are uniform in the block lengths and the number of blocks.

Proof. Set $T_j = -H_j$ and write

$$T_j(w, p) = \sum_{n \neq 0} t_{j,n}(p) e^{2\pi i n w}.$$

The coefficients are holomorphic in p . Denote the negative and positive parts by T_j^- and T_j^+ . Starting from (w_j, p) , continue the coordinates by

$$w_{s+1} = w_s + a_s(p)$$

in either direction, and define

$$G_j(w_j, p) = \sum_{l \geq j} T_l^-(w_l, p) - \sum_{l < j} T_l^+(w_l, p). \tag{6.4}$$

For negative coefficients use a horizontal coefficient-bound line at height $L_l + s - \delta/4$. An evaluation point of $D_j(s - \delta, \rho)$ is below that line, in the continued l -th coordinate, by at least

$$3\delta/4 + (1 - \kappa)(l - j) \quad (l \geq j).$$

For positive coefficients use height $L_l - s + \delta/4$. The evaluation point is above that line by at least

$$3\delta/4 + (1 - \kappa)(j - 1 - l) \quad (l < j).$$

For the nearest terms these inequalities are exactly the two defining inequalities for D_j . For each further term an intervening block contributes at least $1 - \kappa$.

The absolute value of a Fourier mode is $|e^{2\pi i n w}| = e^{-2\pi n Y}$. Therefore both sums, including the sum over frequencies, are bounded by a constant times

$$e \sum_{n \geq 1} \sum_{q \geq 0} \exp(-2\pi n(3\delta/4 + (1 - \kappa)q)) \leq C e \delta^{-1}.$$

This proves normal convergence and holomorphicity, including in the parameter. Telescoping Equation (6.4) leaves $T_j^- + T_j^+ = -H_j$, proving Equation (6.3). Cauchy estimates on smaller D_j give the derivative bounds. In particular, one need not differentiate a sum of an unbounded number of translation functions term by term. $\square$

6.3 The nonlinear iteration

We record the domain issues in the Newton step. Reserve a fixed number of successive margin losses, each comparable to δ , for the primitive, Fourier splitting, derivative estimates, flows, and composition. Enlarging this fixed number if necessary does not depend on any block length. In the following statements the total loss is denoted by $c_0 \delta$, with c_0 a fixed constant.

Apply Lemma 6.2 to $E_j = J_j^{-1} F_j$, and then Lemma 6.3. Let K_j be the time-one map of X_{G_j} . It is defined on the reserved smaller D_j provided $C\delta^{-2}e < \delta$. Its displacement and first-order remainder satisfy

$$\|K_j - \text{id}\| \leq C\delta^{-2}e, \quad \|K_j - \text{id} - X_{G_j}\| \leq C\delta^{-5}e^2. \quad (6.5)$$

Indeed the second estimate follows by integrating the differential equation for the flow and using $\|DX_{G_j}\| \|X_{G_j}\| \leq C\delta^{-5}e^2$. Negative-time flows satisfy the same estimates on the corresponding smaller domains.

The new transition and its proposed model are

$$F_j^{\text{new}} = K_{j+1}^{-1} F_j K_j, \quad J_j^{\text{new}}(w, p) = (w + a_j(p) + u_{j,0}(p), p).$$

To check the cancellation without differentiating a long coordinate change, put $Q_j = G_{j+1} \circ J_j$. The conjugate $J_j^{-1} K_{j+1} J_j$ is the time-one Hamiltonian flow of Q_j . On a buffered seam $\|Q_j\| \leq C\delta^{-1}e$; the same Cauchy and flow estimates apply to it. The linear term of

$$J_j^{-1} K_{j+1}^{-1} J_j E_j K_j$$

is

$$(u_j, v_j) + X_{G_j} - X_{Q_j} = (u_{j,0}, 0) + R_j.$$

Here the Hamiltonian terms cancel by Equation (6.3). The remaining products are quadratic: Equation (6.5) bounds the flow remainders by $C\delta^{-5}e^2$, and the other composition errors are bounded by a first derivative times a displacement. For example, they are bounded by

$$C(\delta^{-1}e)(\delta^{-2}e) \quad \text{or} \quad C(\delta^{-3}e)(\delta^{-2}e).$$

Finally compose with $(w, p) \mapsto (w - u_{j,0}(p), p)$ to normalize by J_j^{new} . Its derivative differs from the identity by at most $C\delta^{-1}e$ and introduces only another quadratic term. It follows, for some fixed integer $q \geq 6$, that

$$\sup_j \|(J_j^{\text{new}})^{-1} F_j^{\text{new}} - \text{id}\| \leq C\delta^{-q}e^2. \quad (6.6)$$

One can use the displayed estimates with $q = 6$; allowing a larger fixed q absorbs harmless reallocations of the margins. No operation in this step involves more than two adjacent charts.

The new left boundary of D_j changes by at most $\|u_{j-1,0}\| \leq e$. Flow displacements are $O(\delta^{-2}e)$. Hence all new domains, translated seams and inverse-flow domains fit in their reserved predecessors when $C\delta^{-q}e$ is sufficiently small compared with δ . The model derivative changes by at most $C\delta^{-1}e$. These observations justify both the domain inclusions used above and their uniform repetition.

Exactness persists. With $\lambda = p dw$, Cartan's formula for our Hamiltonian convention gives

$$\mathcal{L}_{X_H} \lambda = d(pH_p - H).$$

Thus the flows are exact. The shear J_j is exact because $J_j^* \lambda - \lambda = pa'_j(p) dp$ has a holomorphic primitive on the parameter disc. Compositions and inverses of exact maps are exact.

Proof of Theorem 6.1. Start with slightly smaller $s_0 < h$ and $\rho_0 < r$. Fix a small $d_0 > 0$ and take $\delta_n = d_0 2^{-n}$. Choose d_0 so that

$$c_0 \sum_{n \geq 0} \delta_n < \frac{1}{4} \min(s_0, \rho_0).$$

At stage n reduce both margins by $c_0 \delta_n$. Thus their limits remain positive. Set $B = Cd_0^{-q}$, increasing the uniform constant in Equation (6.6) once if needed. If e_n bounds the normalized errors, then

$$e_{n+1} \leq B2^{qn} e_n^2.$$

For $x_n = B2^{q(n+1)} e_n$ this becomes $x_{n+1} \leq x_n^2$. Consequently, if $x_0 \leq \theta < 1$, then

$$e_n \leq B^{-1} 2^{-q(n+1)} \theta^{2^n}.$$

Every series $\sum_n \delta_n^{-k} e_n$, with fixed k , converges and tends to zero as $e_0 \rightarrow 0$. Taking e_0 still smaller ensures all the margin conditions above, keeps $|\text{Im } a_j^{(n)} + L_j| < h/3$, and keeps the translation derivative bound below $A + 1$. These estimates are uniform in j .

The successive compositions of the K_j converge uniformly, with derivatives on smaller domains, to maps C_j into the original block domains. To see this directly, sum their displacements; their derivative norms are bounded by a convergent product of $1 + C\delta_n^{-3}e_n$. The reserved inclusions keep every composition in its domain. The limits remain symplectic because their differentials converge and each determinant is one. Put $\tilde{\phi}_j = \phi_j \circ C_j$ and $\tilde{a}_j = \lim_n a_j^{(n)}$. Passing to the limit in the chart identities on the seams gives

$$\tilde{\phi}_j(w, p) = \tilde{\phi}_{j+1}(w + \tilde{a}_j(p), p).$$

The parameter radius has been decreased by the fixed sum of reserved losses and is still bounded below by a positive number depending only on the initial margins. A smaller common radius b can therefore be fixed before the error threshold, independently of the chain.

It remains to identify the domain, rather than invoke a uniformization theorem. Fix block zero, put $w_0 = w$, and define $w_{j+1} = w_j + \tilde{a}_j(p)$ in both directions. These are holomorphic shears, and $dw_j \wedge dp = dw \wedge dp$. Use the j -th chart between the heights

$$\text{Im } w_{j-1} = L_{j-1}, \quad \text{Im } w_j = L_j,$$

including the surviving overlaps. In w_j coordinates the first height is near zero and the second is L_j . Their separation is at least $1 - h/3$. Hence the successive edges tend to both infinities, locally uniformly in p , and the charts cover all of $\mathbb{C} \times \Delta_b$. They agree on overlaps by the limiting identity. Their common pullback form is $dw \wedge dp$, proving immersion. Periodicity follows from the periodicity of the original charts and every correction. The summable estimates also prove the claimed approximation statements. $\square$

6.4 Enforcing exactness by recentering

The geometric applications supply nearly matching blocks, but not initially exact transitions. The following observation explains precisely how their centers are allowed to move.

Lemma 6.4 (Period adjustment). *Suppose symplectically normalized charts, together with corresponding model charts, are available for centers varying in larger transverse discs. Suppose the actual charts are uniformly close to their models, and transitions at matching model positions are uniformly close to degree-one shears. The transition into a newly added block can be made exact by a transverse translation of that block of size bounded by its transition error. Recenter its model by the same translation. The approximation of that block by its own recentered model retains its original bound.*

Consequently, such adjustments can be repeated indefinitely whenever the updated centers remain in a region with uniform buffered chart choices. No bound on their accumulated displacement relative to the first center is required.

Proof. For a transition $F = (W, P)$ the closed one-form $P dW - p dw$ has a constant complex period

$$D = \int (P dW - p dw).$$

It is independent of the representative period circle and of p . Evaluation at an interior circle with $p = 0$, using Cauchy estimates on a fixed seam margin, bounds $|D|$ by the error from a shear. Replace the added target chart by $(W, P_{\text{new}}) \mapsto \phi(W, P_{\text{new}} + D)$. The transition now has $P_{\text{new}} = P - D$, so its period is

$$D - D \int dW = 0.$$

The last integral is one because the transition has degree one. In a backward extension one instead translates the newly added input chart, with the corresponding sign. Uniform buffers permit these small translations.

Apply the identical substitution to the model chart. Its outgoing end is now the outgoing end of that updated model, and this is the position used to choose the following block. Both actual and model maps were defined on a larger disc, so their difference after restriction and translation has the same uniform bound. The old seam still has only a small error; the assertion does not require the two moved model centers to coincide there. Repeating this argument compares each new chart with its current model and never sums the previous matching errors. $\square$

For completeness, the normalization used in this lemma is compatible with periodicity. If a nearby chart pulls back the surface form to $k(w, t) dw \wedge dt$, with k periodic and close to one, put

$$p(w, t) = \int_0^t k(w, s) ds.$$

Then $dw \wedge dp = k dw \wedge dt$, and $(w, t) \mapsto (w, p)$ is a small periodic biholomorphic change on smaller domains. Its inverse is obtained uniformly with margins. Additive origins in w may also be changed by holomorphic functions of p ; these changes are exact shears, so they do not affect the period adjustment.

7 An open region of K3 periods

Theorem 6.1 permits the centers of successive annuli to move. We now construct a degeneration for which every possible new center admits another block. This is the reason an infinite chain can be continued after the exactness adjustments.

Proposition 7.1. *There is a nonempty open subset of the full marked K3 period domain such that every surface with period in this subset has property L at every point.*

The openness here is unrestricted: the nearby surfaces need not be projective. We shall first produce one smooth quartic with a finite collection of buffered chart constructions, then deform all those constructions to its full local deformation space.

7.1 The semistable model and its regular pencils

Choose two general smooth quadrics $Q_+, Q_- \subset \mathbb{P}^3$. Their intersection E is a smooth elliptic quartic. For a general quartic form G consider the family

$$Q_+ Q_- = \alpha G. \quad (7.1)$$

Here the same symbols Q_+, Q_- denote their quadratic equations. The sixteen zeros Z of $G|_E$ are simple. Near such a zero the total space has equation $xy = \alpha t$, up to holomorphic changes of coordinates. Blowing up one component, locally the ideal (x, α) , is a small resolution. For example, in one chart write $\alpha = xa$; the strict transform has $y = at$ and the map to the parameter disc is $\alpha = xa$. In the other chart write $x = ab$; the equation becomes $t = by$, with smooth coordinates (α, b, y) and projection α . Thus the resolved total space is smooth and the central fiber has normal crossings.

Its two components, with a suitable choice of labels, are a quadric S^+ and the blowup S^- of a quadric at Z ; they meet along the strict transform of E , again denoted E . The resolution is crepant because it is small and the original hypersurface is Gorenstein. Adjunction for the quartic family supplies a nowhere-zero relative dualizing form. On $S^\pm$ it becomes a logarithmic symplectic form $\omega_\pm$ with simple pole along E and opposite nonzero residues. In a normal-crossing chart we have

$$uv = \alpha, \quad \omega_\alpha = k(u, v, t) \frac{du}{u} \wedge dt, \quad (7.2)$$

where k is holomorphic and nowhere zero and t is a coordinate along E . In particular $k(0, 0, t)dt$ is its residue on the u -component; the residue on the v -component is its negative. For small nonzero α , the fibers of Equation (7.1) are smooth quartic K3 surfaces.

Call a fiber of a base-point-free pencil on S^+ or S^- *regular* if it is a smooth rational curve, the pencil map is smooth along it, and it meets E transversely at two distinct points. We use the two ruling pencils, pulled back on S^- , and on S^- also the pencils of $(1, 1)$ curves through two distinct points of Z , with their two base points subtracted.

Lemma 7.2. *The quadrics and G can be chosen so that:*

1. *Through every point of E on each component there is a regular fiber of one of these finitely many pencils.*
2. *The union of their regular fibers contains the complement of a proper algebraic subset of each component.*
3. *Every regular fiber has, away from E , a transverse intersection with a regular fiber of another pencil.*

Proof. On a plain quadric, identify the two rulings with the projections of $\mathbb{P}^1 \times \mathbb{P}^1$. Their restrictions to E have degree two. They cannot both ramify at one point: otherwise the differential of the inclusion of the smooth curve E in the quadric would vanish. This proves the first assertion on S^+ .

For the blown-up quadric, a ruling fiber is additionally forbidden when it passes through a point of Z . Choose Z generally so that its points avoid both ruling ramification sets, and so that a forbidden level of one ruling and a forbidden level of the other have a common point on E only at a point of Z . To check that these are proper conditions, write the two projections as π_1, π_2 . Their ramification sets are disjoint. Conditions involving one ramification set and one $\pi_i(Z)$ exclude only finitely many values for the corresponding point of Z . For distinct z_i, z_j , the condition that the grid point with coordinates $(\pi_1(z_i), \pi_2(z_j))$ lie on E is a proper condition on the pair. For $i = j$ the grid point is z_i itself. Thus, away from Z , at least one ruling remains regular.

Fix $z_i \in Z$. Choose another center z_j and take the plane through the tangent line to E at z_i and through z_j . For general choices its intersection with the quadric is a smooth $(1, 1)$ curve. Its divisor on E is

$$2z_i + z_j + z_k,$$

with z_k different from z_i, z_j , and it passes through no other point of Z . Its strict transform is a member of the pencil of $(1, 1)$ curves through z_i, z_j . The chord through these two points is not a ruling line; hence the pencil has exactly these two simple base points, which are resolved by the blowups. An order-two contact at z_i becomes a simple transverse intersection with the strict transform of E . The simple intersection at z_j disappears, and the intersection at z_k remains. The selected strict transform is therefore a regular fiber through the point of E over z_i .

All the generic requirements just used are simultaneous finite proper conditions. More explicitly, on the elliptic curve the residual z_k is determined by the hyperplane divisor class and $2z_i + z_j$; excluding coincidences or another center imposes a proper condition on at most three of the centers. The other requirements involve at most two centers and fixed branch data. The divisor Z ranges over the complete degree-sixteen linear system $|\mathcal{O}_E(4)|$. Indeed the restriction of quartic forms is surjective, as follows from the Koszul resolution for the intersection of two quadrics. In an ordered general divisor of this degree any specified fewer than sixteen points vary freely; the remaining points satisfy the one elliptic sum condition. Hence the finitely many proper conditions above can all be avoided.

For each of the finitely many pencils, only finitely many members fail to be smooth or meet E nontransversely. Their union is a proper algebraic set, proving the second assertion. Finally, on any regular rational fiber, at least one of the two ruling projections is nonconstant. This is immediate for a ruling fiber by using the other ruling; for a $(1, 1)$ fiber both projections are nonconstant. Away from finitely many points of the fiber the corresponding ruling member is regular and the projection has nonzero derivative. Such an intersection is transverse and can be chosen off E . $\square$

We now make finite compact choices, each with an open buffer. Let G_{reg} be the union of *all* regular fibers from the listed pencils. Its complement is contained in a proper algebraic set, and

G_{reg} contains a neighborhood of E . By the first assertion of Lemma 7.2, compact subfamilies of regular fibers supply entries at every point of E . By its third assertion each required family can, after restriction, be equipped with a transverse partner at an interior point, with a positive transversality margin in local charts. Compactness supplies finitely many such choices near E .

For each point of the algebraic exceptional set away from E , choose a coordinate line not contained in that set and a small radius whose boundary avoids it. The boundary then lies in G_{reg} . Small translations and tilts retain this boundary containment, so the disc construction supplies a neighborhood of possible centers. Points of G_{reg} themselves have regular coordinate neighborhoods. On the compact region outside a chosen neighborhood of E , finitely many of these two kinds of center neighborhoods suffice. Only after choosing these finitely many discs do we add finitely many regular patches covering their compact boundaries, along with the corresponding regular families and transverse partners. Thus every retained datum belongs to a finite buffered collection. The algebraic exceptional set is used before the finite selection. These disc choices will be used with Corollary 2.10 after constructing the strips.

7.2 Core and neck coordinates

Lemma 7.3. *On a small parameter disc of any compact regular family there are coordinates $(z, p) \in \mathbb{P}^1 \times \Delta_r$ in which the two ends are $z = 0, \infty$ and*

$$\omega_{\pm} = \frac{1}{2\pi i} \frac{dz}{z} \wedge dp.$$

Either end can be designated as the entrance. With entrance at ∞ , exit at 0, and $z = c \exp(2\pi i w)$, this reads $\omega_{\pm} = dw \wedge dp$. The parameter p is a local affine coordinate for either end position on E , read using its signed residue differential.

Proof. Over a small disc the smooth proper family of rational fibers is a holomorphic $\mathbb{P}^1$ bundle and is trivial there. One can see this by choosing a local section, taking its relative degree-one line bundle, and using sections and base change to identify the family with a projective bundle. The two distinct end sections can then be put at zero and infinity.

In a parameter s on this disc write the logarithmic form as $f(z, s)dz \wedge ds$. For each s , the one-form $f(z, s)dz$ on $\mathbb{P}^1$ has only simple poles at zero and infinity. It is therefore $b(s)dz/z$. Nondegeneracy makes b nonzero. Set $dp = 2\pi i b(s)ds$.

Let $t_+(s)$ and $t_-(s)$ be the exit and entrance positions on E , and let ρ be the residue differential on this component. Taking residues gives the exact identities

$$dp = 2\pi i t_+^* \rho = -2\pi i t_-^* \rho.$$

They also follow from the constancy of the elliptic sum of the two intersection points. Since both ends are transverse, t_+ and t_- are locally invertible. Thus p moves each end freely and the stated residue interpretation follows. $\square$

In a fixed normal-crossing chart at an end, its nonzero normal coordinate on the corresponding central component has the uniform expansions

$$u = k_-(p)z^{-1}(1 + O(z^{-1})) \quad \text{at the entrance,} \quad (7.3)$$

$$u = k_+(p)z(1 + O(z)) \quad \text{at the exit.} \quad (7.4)$$

The coordinate along E differs from its end value by $O(z^{-1})$ or $O(z)$ respectively. The functions $k_{\pm}$ are nonzero, with uniformly bounded logarithmic derivatives on the selected smaller parameter discs.

These estimates, with any fixed number of derivatives in logarithmic and parameter coordinates, are uniform over the finite buffered atlas.

Choose a small positive normal cutoff d . Truncate a core at its two ends where $|u|$ is comparable to d , retaining fixed buffers in logarithmic height. In the coordinate w of Lemma 7.3, this is a cylinder of length tending to infinity as $d \rightarrow 0$, uniformly over the compact families. For every fixed d it is a relatively compact annulus times a disc in the smooth part of the central fiber. Lemma 2.7 deforms this map into nearby smooth fibers on a slightly larger annulus and parameter disc. Its symplectic normalization gives maps with pullback form $dw \wedge dp$ converging to the central maps on these fixed domains. This uses only finite-domain deformations, and can be done simultaneously for the finite raw atlas. Restrictions and recentering of its larger parameter discs then supply all the moving centers used below.

The connecting neck has a more direct description. In Equation (7.2) put

$$u = c \exp(2\pi i w), \quad v = \alpha/u,$$

and allow it to run from $|u| \asymp d$ to $|v| \asymp d$, with fixed additional logarithmic buffers. Normalize its transverse coordinate by

$$p(w, t) = \int_{t_0}^t 2\pi i k(u, \alpha/u, s) ds. \quad (7.5)$$

This gives $dw \wedge dp = \omega_\alpha$. On this entire neck, including its buffers, both $|u|$ and $|v|$ are at most Cd . Consequently

$$p(w, t) = \int_{t_0}^t 2\pi i k(0, 0, s) ds + O(d).$$

The estimate is uniform in neck length, and holds with any fixed number of derivatives on smaller parameter discs and logarithmic buffers. Indeed each w derivative differentiates u or v by a fixed constant multiple, and both remain $O(d)$; parameter derivatives are bounded on the fixed t -patches. The derivative in t is bounded away from zero, so the normalization is uniformly invertible there.

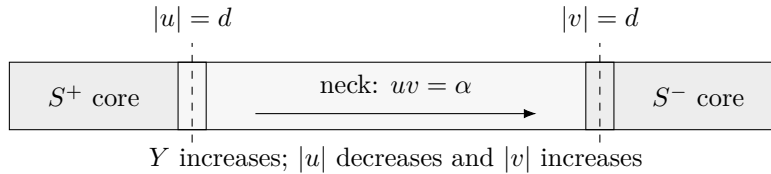


Figure 3: The core–neck joins in logarithmic height, not to scale. The overlap widths are fixed. Once d is fixed, the neck length grows as $\alpha \rightarrow 0$, while its transverse width stays bounded below. After sewing, the same parameter p labels a leaf across both joins.

Figure 3 separates the fixed-width overlaps from the part of the cylinder whose length grows. Only the former enter the local transition estimates.

Lemma 7.4 (Buffered seams). *There are fixed positive parameter and seam widths and a fixed bound on translation derivatives with the following property. Choose d sufficiently small, then $\alpha \neq 0$ sufficiently small depending on d . At matching nominal end positions the normalized core and neck charts have degree-one symplectic transitions of the form*

$$F_j = J_j \circ (\text{id} + E_j), \quad J_j(w, p) = (w + a_j(p), p),$$

on buffered seams. After choosing long-coordinate origins and cut heights, they satisfy the domain hypotheses of Theorem 6.1, apart from exactness, and

$$\sup_j \|E_j\| \leq Cd + \epsilon(d, \alpha), \quad \epsilon(d, \alpha) \longrightarrow 0 \quad (\alpha \rightarrow 0, d \text{ fixed}).$$

The widths and the bound for a'_j can be chosen before d .

Proof. At an exit seam Equation (7.4) has leading part $u = k_+(p)z$. Taking a local logarithm of the nonzero coefficient gives a change $w_{\text{neck}} = w_{\text{core}} + a(p)$. Its derivative is bounded by the logarithmic derivative of k_+ . Its transverse change is exactly the identity at leading order: both parameters are primitives of the same signed residue differential, with the same constant at the matching nominal end.

At the entrance into the next component, the leading relation is $v = k_-(p')/z'$ and $wv = \alpha$. Hence z' is a nonzero holomorphic multiple of u , so again $w' = w + a(p)$, with the same orientation. The sign of the residue changes between the two components, and the use of $z' = \infty$ at the entrance changes it once more. The transverse primitives therefore again agree as germs. The potentially large constant $\log \alpha$ affects the translation itself, not its derivative in p .

The actual seam maps differ from these leading maps by $O(d)$ from the expansions and Equation (7.5), plus the core deformation error $\epsilon(d, \alpha)$. These estimates are in $(\log u, t)$ or $(\log v, t)$ coordinates. The leading changes are uniformly invertible there. The inverse function theorem with the reserved buffers therefore produces the actual inverse branches throughout the seams. Local branches agree by uniqueness near the specified leading branch, including after going once around the period. Their lifts have degree one, and they are symplectic because both charts pull back the same surface form to $dw \wedge dp$.

For completeness, the cuts can be prepared without imposing incompatible origins at the two ends of a block. Work first in raw coordinates. Enlarge each core range to $cd < |z| < 1/(cd)$ with a fixed sufficiently small $c > 0$, and add fixed logarithmic buffers to every neck. At a seam choose the two matching cut heights by its leading formula at the nominal center. Each raw block has independent room for the choice of its two cuts. Then translate its single long-coordinate origin so that its entrance is at height zero; its exit defines L_j . For small d , and then small α , all core and neck lengths are at least one. Shrink the fixed transverse discs so that the variation of $\text{Im } a_j$ about its central value is less than $h/8$. The logarithmic derivative bounds show that this shrinking is independent of d and α . The two cut heights then give precisely $|\text{Im } a_j + L_j| < h/8$ with the required margins. $\square$

7.3 Infinite continuation and transverse partners

We spell out the compactness statement needed for an infinite construction. The finite atlas has larger and smaller end-position patches whose interiors cover E . At a designated end choose a regular family and a neck patch whose smaller patches contain it. Their larger patches give a uniform positive buffer, after taking a finite refinement if necessary. Thus a transverse translation whose size is less than a fixed threshold still belongs to the chosen larger patches, for every designated position on E . Both orderings of the ends of a core are included in the atlas. If a later choice requires a smaller positive working radius, use more centers from these same compact families. Their smaller end-position neighborhoods again cover E , and a finite subcover can be chosen. These are restrictions and translations of the same raw charts, so the uniform derivative and buffer bounds do not change. In particular, the positive parameter width retained by the sewing theorem can be fixed before the cutoff d without losing coverage.

Start with any prescribed core or neck chart. Match its outgoing nominal end with a new chart. By Lemma 7.4 its period defect is $O(d + \epsilon(d, \alpha))$. Apply Lemma 6.4 to make the seam exact, translating the actual and model transverse coordinates of the newly added chart together. Use that updated model's outgoing end for the next choice. It is some point of E , where another uniformly buffered choice is available. Continue alternately through cores on the two components and intervening necks. In the opposite direction perform the corresponding adjustment on each newly added input block. The initial block need not be moved in either construction.

The displacement of the nominal centers over many steps can be large. This causes no loss of the estimates: at each step the comparison is with that block's own translated model on a larger prepared patch, and the outgoing model position is updated before the following seam is formed. The set of possible positions is the whole compact curve E , not a single fixed local parameter disc. The actual parameter disc in each centered chart, in contrast, has the same fixed radius. This is exactly the distinction used in Lemma 6.4.

Take d small and then α small enough that all seam errors, including the period adjustments, are below the threshold of Theorem 6.1. We obtain periodic symplectic immersions of $\mathbb{C} \times \Delta_b$ whose corrected charts are uniformly close to the corresponding raw charts. By making the initial errors smaller, their first derivatives on the fixed interior core patches are as close as required.

At a designated interior crossing use an additional chain starting from the transverse partner pencil. In the central model the two tangent directions have a fixed positive angle in the chosen local coordinates. The two corrected families still have independent directions there, and the partner chart still covers a smaller neighborhood of the crossing. For every sufficiently small transverse parameter on the first strip, a point on its leaf in this core patch therefore lies on a leaf of the partner family, with transverse intersection. This uses the partner *family*: no interpolation at a prescribed point is needed.

By Remark 2.5 and Proposition 2.2, these strips imply property L at every covered point on the first strip, including points in other blocks of its chain. The unbounded coordinate may be translated to the crossing, and the common parameter p identifies the same leaf all along the chain. A strip initialized in a neck reaches one of the adjacent prescribed regular core families and uses its chosen partner in exactly this fashion.

7.4 Coverage and unrestricted deformation

There are two coverage issues: the middle of an increasingly long neck and the neighborhood of either cutoff. Neither may be discarded as a small set. We verify them in coordinates with uniform bounds.

Inside a core away from its ends, the deformed chart and its sewing correction are close in C^1 to the central regular chart. Local invertibility with a margin implies that its image contains every prescribed smaller regular patch, under a local identification of the smooth fibers. This includes the compact boundary patches of all the coordinate discs chosen earlier. The same statement is uniform over compact subfamilies and small recenterings.

For a neck, the normalized raw map is already a coordinate chart in $(\log u, t)$ on the whole buffered interval from $|u| \asymp d$ to $|v| \asymp d$. The correction supplied by Theorem 6.1 is uniformly close to the identity in (w, p) and is defined with fixed margins, independently of the interval length. Its image contains a uniformly smaller source cylinder: at any point of that smaller cylinder, a fixed-radius coordinate ball lies in the larger one, and the small-change inverse theorem applies there. Thus the corrected chart covers the entire smaller neck interval and the smaller end-position patch, not only a compact part chosen in advance.

At the u -cut use a normal-crossing patch about a point of E . On the component side, Equations (7.3) and (7.4) give uniformly nonsingular leading coordinates $(\log u, t)$, with error $O(|u|)$. The coordinate t differs from the endpoint coordinate by $O(|u|)$. Choose a fixed small normal size $d_* > 0$, smaller than a fixed fraction of the available end-position width. The selected core charts cover the region from a fixed multiple of d up to d_* , after shrinking their end-position patches. Their sewing corrections are uniformly small in the same logarithmic coordinates. The neck chart covers below that multiple of d and overlaps the core region by its extra logarithmic buffers. The identical argument applies at the v -cut. A finite collection of the smaller end-position patches covers E .

This argument does not ask for a lower bound on an ordinary metric chart radius at $|u| = d$. All inverse estimates there are in logarithmic and end-position coordinates, where the leading maps and the buffers have uniform bounds. One first chooses the position widths and the needed error bound for coverage and for the core crossings, and then chooses d . The fixed d_* can be reduced after the widths are chosen; finally take d much smaller than d_* . On the remaining compact region outside these end neighborhoods, the preceding finite collection of regular patches and coordinate discs applies. If the end neighborhoods have been reduced in choosing d_* , the intervening closed annular region is already covered by the same regular core families; it is a fixed compact region before d is chosen. Properness of the semistable family ensures that these core and neck regions cover every nearby smooth fiber.

At each reserved coordinate disc away from E , use the submersion coordinates of the total space to transport its local coordinate construction to the smooth fibers. Its boundary, and the boundaries of sufficiently nearby translated or tilted discs, remain in the regular patches where L was just proved. Corollary 2.10 gives L throughout the remaining smaller coordinate neighborhoods. Thus L holds at every point of the selected smooth quartic.

Now fix the small nonzero α used above. Every raw core and neck chart is a map from a *finite* annulus times a disc, with margins. There are finitely many raw charts; all the moving versions are restrictions and translations of these larger charts. Apply Lemma 2.7 in the full local deformation family of this smooth K3, choosing a nearby holomorphic symplectic form and normalizing each chart. They deform simultaneously with arbitrarily small additional error. Their seam inverse branches persist: on each compact buffered seam they are the nearby branches of the original local inverse, and uniqueness makes those branches agree throughout the seam and around its period circle. Thus all the domain, derivative, period-adjustment, coverage, and transversality estimates persist in a sufficiently small unrestricted deformation neighborhood. This argument does not deform the infinitely many corrected charts individually; it deforms the finite raw atlas and repeats its uniformly controlled construction.

The order of choices is worth recording because it prevents any dependence of the allowable error on the length of the neck:

1. Choose the finite compact regular families, partner crossings, end-position patches, disc boundaries, parameter widths and logarithmic buffers.
2. Choose an allowable sewing and coverage error for these data; the derivative bounds in the leading shears have already been fixed.
3. Choose d so that the $O(d)$ errors are below that allowance, and then choose a nonzero α so that all fixed- d deformation errors are below it.
4. Choose the full deformation neighborhood of that fixed smooth quartic so that its additional errors remain within the same allowance.

K3 deformation theory gives a smooth local deformation space, and the marked period map is a local isomorphism by local Torelli; see [24]. Its image of this last neighborhood is a nonempty open subset of the full period domain. Any other marked K3 with the same period has an isomorphic underlying surface: signs and reflections in algebraic roots match the Kähler chambers while fixing the period, and global Torelli then applies [24]. Consequently every surface with a period in this open subset has L everywhere, proving Proposition 7.1. Theorem 3.1 consequently gives convex approximation for each of these surfaces.

8 Periods containing a rational direction

Let

$$\Lambda = 3U \oplus 2E_8(-1), \quad \mathcal{D} = \{P \subset \Lambda \otimes \mathbb{R} : P \text{ is an oriented positive two-plane}\}.$$

Here positivity refers to the intersection form, denoted by $(\ , \)$. For a primitive vector $v \in \Lambda$ with $(v, v) > 0$, set

$$\mathcal{D}_v = \{P \in \mathcal{D} : v \in P\}.$$

All open subsets of $\mathcal{D}_v$ below have its relative real topology. We use the K3 period, Torelli, and lattice theorems in their usual unpolarized form; a reference for these results and the facts about genus-one pencils used below is [24, Chapters 2, 6–8, and 11].

Theorem 8.1. *For every primitive positive integral vector $v \in \Lambda$, there is a nonempty relatively open subset $\mathcal{O}_v \subset \mathcal{D}_v$ such that every marked K3 surface with period in $\mathcal{O}_v$ has property L at every point.*

On a marked K3 surface whose period contains v , the holomorphic symplectic form has a unique normalization satisfying

$$[\mathrm{Im} \sigma] = v. \tag{8.1}$$

Indeed the real and imaginary parts of a nonzero holomorphic two-form are orthogonal and of equal positive square, and multiplication by a nonzero complex scalar gives all oriented orthogonal frames of its period plane. This normalization varies smoothly on $\mathcal{D}_v$.

We first produce a convenient central surface, then construct strips on its nearby deformations satisfying (8.1).

8.1 An isotrivial center for every positive integral vector

Lemma 8.2. *For every primitive $v \in \Lambda$ of square $2d > 0$, there is a marked projective K3 surface X_0 whose period contains v and which admits a genus-one fibration*

$$\pi : X_0 \longrightarrow \mathbb{P}^1$$

with all smooth fibers of j -invariant zero. Its primitive fiber class $e \in \Lambda$ satisfies $(e, v) = 0$.

Proof. We begin with an auxiliary elliptic K3 surface with section, given by the Weierstrass equation

$$y^2 = x^3 + a(t),$$

where a is a section of $\mathcal{O}_{\mathbb{P}^1}(12)$ with twelve simple zeros, and the fundamental line bundle is $\mathcal{O}_{\mathbb{P}^1}(2)$. The canonical bundle formula gives trivial canonical bundle, and $R^1\pi_*\mathcal{O} = \mathcal{O}_{\mathbb{P}^1}(-2)$ gives irregularity zero. The simple zeros give cuspidal fibers with smooth total space. Thus the smooth

Weierstrass surface is a K3 surface. These global Weierstrass and direct-image formulas are recalled in [24, Chapter 11, Sections 1–2].

Let g be the automorphism which multiplies x by a primitive cube root of unity. Its fixed locus is the disjoint union of the zero section and the double cover $x = 0$ of the base. The latter is branched at twelve points and has genus five. The fixed locus has Euler characteristic $2 + (2 - 10) = -6$. The topological Lefschetz formula, in the order-three K3 formulation of [2, proof of Theorem 2.2], gives

$$\mathrm{tr}(g \mid H^2) = -8.$$

If r is the invariant rank, the other eigenvalues occur in conjugate pairs of primitive cube roots, so $r - (22 - r)/2 = -8$, whence $r = 2$. The fiber and zero section generate a unimodular hyperbolic plane in this invariant lattice; consequently the invariant lattice is exactly U . Its orthogonal complement is even unimodular of signature $(2, 18)$ and contains two hyperbolic planes.

That complement contains a primitive vector of square $2d$. Eichler's criterion, applied to the unimodular K3 lattice with its two hyperbolic summands, says that primitive vectors with the same square are isometric [20]. We may therefore transport the preceding lattice isometry so that the specified vector v belongs to its noninvariant complement. There $1 + g + g^2 = 0$, and thus

$$(v, v) = (gv, gv) = 2d, \quad (v, gv) = -d.$$

The plane $P_0 = \mathrm{span}_{\mathbb{R}}\{v, gv\}$ is positive. Choose either of its two orientations, and realize it as a marked K3 period by period surjectivity. The invariant U is algebraic at this period and contains a positive integral class, so the resulting surface is projective.

Choose a primitive isotropic generator e of this U , with sign chosen so that its represented class is effective. We keep the abstract lattice data g, U, e fixed and change the marking by reflections in algebraic (-2) -classes so that the class represented by e becomes nef. The reflections fix P_0 pointwise, hence fix v ; all preceding lattice relations are unchanged. For completeness, if an effective isotropic class has negative intersection with an irreducible curve, that curve is a (-2) -curve; reflection in it reduces the positive integral degree against a fixed ample class. Repetition terminates in the nef cone. The primitive nef isotropic pencil theorem then gives a base-point-free complete pencil $|e|$ with

$$h^0(\mathcal{O}(e)) = 2, \quad H^1(\mathcal{O}(e)) = 0.$$

It defines a genus-one fibration on our center.

We must show that this particular pencil is isotrivial. The g -invariant period planes of the chosen eigenvalue form a complex ball: they are the positive lines in one of the two nonreal eigenspaces of g on $U^\perp \otimes \mathbb{C}$. Take a small ball through P_0 , realized by a marked local K3 family. At a generic point its algebraic lattice is precisely U . To see this, each integral class outside U imposes a proper linear Hodge condition; such a condition cannot vanish on the entire eigenspace because the two conjugate eigenspaces span $U^\perp \otimes \mathbb{C}$. Their countable union has empty interior.

The line bundle with class e extends to this small family. One way to verify the extension is to take a contractible Stein base. The topological class extends to the total space and has zero image in $H^2(\mathcal{O})$: by the Leray spectral sequence and Stein vanishing this group is the space of sections of $R^2\pi_*\mathcal{O}$, and the class vanishes fiberwise because it is of type $(1, 1)$. The exponential sequence supplies the line bundle. The vanishing of $H^1(\mathcal{O}(e))$ and cohomology and base change extend its two sections. Their having no common zero is open by properness. Thus the genus-one pencils persist near P_0 .

At every nearby generic period, g acts identically on the Picard lattice, hence fixes an ample class. Global Torelli realizes it as an automorphism of order three. Suppose its induced action on the pencil base were nontrivial. All its fixed points would then lie over the two fixed points of that base action. Every fixed curve would be vertical. Since an irreducible vertical curve C satisfies $(C, e) = 0$, the Hodge index theorem gives $C^2 \leq 0$ and adjunction gives $p_a(C) \leq 1$. The fixed locus of a finite-order automorphism of a smooth surface is a disjoint union of smooth curves and isolated points. It would therefore have nonnegative Euler characteristic, contrary to the value -6 determined by the cohomological action.

The base action is consequently trivial. The multiplier on a smooth fiber's holomorphic differential equals the primitive cube-root multiplier on the surface form. A smooth elliptic curve with such an automorphism has $j = 0$. Finally fix any smooth member of the central pencil. It remains smooth in the extended pencil over a neighborhood in the deformation base, and its j -invariant varies continuously there. The nearby generic values are zero, so the central value is zero as well. This proves the assertion for every smooth member at P_0 . The relation $(e, v) = 0$ holds because e belongs to the original invariant lattice. $\square$

8.2 Actions on the smooth fibers

Fix the surface of Lemma 8.2 and the normalized form σ_0 . Write $S \subset \mathbb{P}^1$ for the finite set of singular values. The linear monodromy of a constant-modulus family of elliptic curves is finite. Choose a connected finite cover of $\mathbb{P}^1 \setminus S$ on which that monodromy is trivial, and compactify it to a smooth compact curve B . Denote the resulting punctured curve by B° . A fixed oriented basis γ_1, γ_2 now identifies the homology of all smooth fibers over B° .

The base change need not possess a global section. Locally it has holomorphic sections and product torus charts. For any primitive cycle γ there are cylinder coordinates, with the chosen cycle represented by the period 1, in which

$$\sigma_0 = dw \wedge dp, \quad dp = \eta_\gamma. \quad (8.2)$$

Here η_γ is a nowhere-zero holomorphic one-form on the corresponding base patch. Indeed, in a local product chart the coefficient of the surface form is constant along each compact elliptic fiber. The period normalization makes these base forms agree on overlaps. They therefore define a global one-form η_γ on B° . For the two basis cycles the forms are in a constant nonreal ratio.

If l is an oriented smooth loop in a K3 surface with normalized form σ , define its action by

$$H_\sigma(l) = \int_C \text{Im } \sigma \pmod{\mathbb{Z}}, \quad \partial C = l. \quad (8.3)$$

The chain C exists because $H_1(X, \mathbb{Z}) = 0$, and the value is independent of C because $[\text{Im } \sigma] = v$ is integral. On X_0 , representatives of a fixed homology class in a smooth elliptic fiber give the same action, since the restriction of σ_0 to that fiber is zero. Let

$$h_\gamma : B^\circ \longrightarrow \mathbb{R}/\mathbb{Z}$$

be this action function. Orientations in (8.2) may be chosen so that $dh_\gamma = \text{Im } \eta_\gamma$; changing all these signs would make no difference below. Actions are additive in the cycle.

Lemma 8.3. *The forms η_γ and functions h_γ extend to B . The map*

$$h = (h_{\gamma_1}, h_{\gamma_2}) : B \longrightarrow (\mathbb{R}/\mathbb{Z})^2$$

has finite fibers. For each primitive γ , put

$$F_\gamma = h_\gamma(B \setminus B^\circ).$$

Outside a finite subset of B , at least two of the three cycles $\gamma_1, \gamma_2, \gamma_1 + \gamma_2$ have action outside their respective finite sets F_γ . Any two of these cycles form an integral homology basis.

Proof. Integration over the elliptic fibers in local product charts expresses the pullback symplectic volume as a fixed positive constant times $i\eta_\gamma \wedge \overline{\eta_\gamma}$. The base change has finite degree and X_0 is compact, so this one-form has finite L^2 norm on B° . In a punctured local coordinate, a holomorphic one-form with finite L^2 norm has no negative Laurent coefficients. It thus extends holomorphically across each puncture. A local primitive of the extended form also extends the circle-valued action function.

The holomorphic forms η_{γ_1} and η_{γ_2} have constant nonreal ratio. Consequently an invertible real linear map identifies the pair of their imaginary parts with a complex-linear differential. Endow $(\mathbb{R}/\mathbb{Z})^2$ with this constant complex structure. Then h is a nonconstant holomorphic map from the compact Riemann surface B to an elliptic curve. It has finite fibers.

A point without two safe choices satisfies two forbidden-level conditions. Each pair of the integral linear functions $h_{\gamma_1}, h_{\gamma_2}, h_{\gamma_1} + h_{\gamma_2}$ is an automorphism of the real two-torus. Thus any two such conditions restrict h to a finite set of torus values. The inverse image is finite by the preceding paragraph. $\square$

We call a cycle safe at a point if its action avoids F_γ . Every closed set of safe action values has compact inverse image in B° . This elementary compactness statement, rather than a bound on the number of cylinder blocks, will control all the successive center adjustments.

8.3 Uniform strips after deformation

Take a marked local deformation of X_0 , identify its underlying smooth fibers, and restrict its parameters to $\mathcal{D}_v$. Let X_t and σ_t denote the fibers and normalized forms. The genus-one fibration is used only on X_0 ; it is not asserted to survive on X_t .

Lemma 8.4. *Fix compact families of regular torus charts on X_0 , each equipped with a safe primitive cycle, and suppose their actions are separated from the corresponding forbidden values by a positive margin. Prescribe any finite number of period cells in these initial data. For all sufficiently small $t \in \mathcal{D}_v$, each datum gives a symplectic immersed strip on X_t , periodic in its long coordinate with period 1, of positive transverse width. On the prescribed initial cells its chart is arbitrarily close in C^1 to the original cylinder chart. All widths and closeness requirements can be chosen uniformly over the given compact data.*

Proof. The centers may move after each exactness correction. We therefore prepare the compact region in which the construction will continue before choosing the charts. Let $\rho > 0$ be a common lower bound for the distance of the specified initial actions from their forbidden sets. For every cycle used in the initial data, put

$$K_\gamma = \{b \in B : \text{dist}(h_\gamma(b), F_\gamma) \geq \rho/4\}.$$

This set is compact in B° . Choose finitely many product-torus charts covering it, with larger buffered domains still lying in B° . All subsequent restrictions and recentering will use this finite atlas. We shall prove that the updated centers remain a fixed distance inside the prepared region.

In a chart for a chosen cycle, the torus has constant lattice $\mathbb{Z} + \tau\mathbb{Z}$, where $\text{Im } \tau > 0$ after choosing a positive complementary cycle. Unroll this complementary period while retaining w modulo $\mathbb{Z}$. We obtain finite cylinders of any prescribed fixed length. Starting sections can be chosen at any fiber phase, so the chart families cover all phases, not just a particular section.

At matching nominal base positions, successive model cylinders are identified on their overlaps by

$$(w, p) \longmapsto (w + a(p), p). \quad (8.4)$$

The functions a incorporate both changes of local section and any required number of complementary periods. The latter are constant, because the modulus is constant. On smaller members of the finite buffered atlas, the derivatives of the remaining translation functions have uniform bounds. Choose long block cores containing the prescribed cells and leave several period cells as buffers at each end. At a joint, integral complementary translations let the two cut heights agree at the center with a bounded adjustment inside these buffers. Both cuts of each raw chart may be chosen independently. After translating the long origins, these charts have the block and overlap geometry required by Theorem 6.1. Shrinking the transverse radius makes the variations of their cut heights uniformly smaller than the overlap margins.

Each cylinder times its transverse disc is Stein. The local deformation argument used earlier therefore deforms these maps on their finite buffered domains to the nearby fibers. Although a cylinder map is usually not injective, it is an immersion, and on each overlap its local inverse branch close to (8.4) is unique. The branches consequently agree around the cylinder, including its period identification. The resulting transition is a degree-one periodic lift. The pulled-back forms can be normalized to $dw \wedge dp$ by the transverse integration used in Theorem 6.1. All these constructions are uniform on the finite atlas, and their errors tend to zero with t .

We next compare the action of an individual deformed block with that of its model, before any blocks have been joined. Let b be the nominal base position of a freshly centered block, and let l_b be the image of a period loop on its zero slice. Uniformly over the prepared charts and their allowed recenterings,

$$\text{dist}_{\mathbb{R}/\mathbb{Z}}(h_\gamma(b), H_{\sigma_t}(l_b)) < \varepsilon_t, \quad \varepsilon_t \longrightarrow 0. \quad (8.5)$$

To verify this, under the fixed smooth identification the real forms $\text{Im } \sigma_t - \text{Im } \sigma_0$ are exact and tend to zero smoothly. Fix a Riemannian metric on the underlying compact manifold and let G be its Hodge Green operator. For the exact two-form $\eta_t = \text{Im } \sigma_t - \text{Im } \sigma_0$, put $\beta_t = d^*G\eta_t$. Its harmonic projection vanishes and $dG\eta_t = Gd\eta_t = 0$, so Hodge decomposition gives $d\beta_t = \eta_t$. Elliptic regularity makes β_t tend to zero smoothly; see [8, Theorem 3.2 and Equation (3.26)] for this Hodge decomposition and Green-operator formulation. Integration of these primitives around the model loops, together with the small cylinders between nearby model and actual loops, proves (8.5). The loop families are compact: the base positions and fiber phases range over compact sets, and the number of wraps in a block is fixed. The estimate depends on the comparison of this block with its own model, not on any previous choices of centers.

Starting from any chosen block, extend the chain in both directions. At a joint initially use a chart with matching nominal base position. Its actual symplectic transition is close to (8.4). Its period defect is the constant

$$\int_{\mathbb{R}/\mathbb{Z}} (P dW - p dw).$$

Moving this candidate's transverse center by the appropriately signed defect makes the period exactly zero. Retain this translated actual chart as the next block, so the exact overlap just obtained is preserved. Move its nominal center by the same amount in the model p coordinate;

that updated position is used only when selecting the following candidate block. The adjustment is small, and comparison with the model of the newly centered chart retains its original error bound. It remains to justify that the nominal positions never approach a puncture.

For this purpose use the action (8.3) on the actual surface. Its value on the period loop of the zero slice is constant within a block. It is also constant between adjacent blocks. Indeed the primitive of $dw \wedge dp$ is $-p dw$; the change of action between the image of a zero-slice loop and the next zero-slice loop is the imaginary part, up to sign, of the displayed period defect. The image loop winds once in the next cylinder. Vanishing of the defect proves equality of its action with that of the next zero slice. Denote this common circle value by c .

Thus the single-block estimate (8.5) reads $\text{dist}_{\mathbb{R}/\mathbb{Z}}(h_\gamma(b), c) < \varepsilon_t$ at every stage. We can now verify that the next center stays in the prepared region. Require the action error and the change in nominal action caused by a single center adjustment to be less than $\rho/8$. For an initial block the conserved value c has distance greater than $7\rho/8$ from F_γ . Inductively, a next nominal center may first be chosen at the preceding one; the small adjustment keeps it inside the prepared region. The exact action comparison then improves its distance from the forbidden set to greater than $3\rho/4$. This restores a fixed margin for the following step. The same argument applies backwards. Thus the entire doubly infinite chain is available with common transverse and overlap margins.

We have obtained exact symplectic transitions with uniformly small errors and the block geometry required by Theorem 6.1. That theorem supplies the periodic strip and arbitrarily small corrections on every prescribed compact part of the initial block. Estimates on slightly larger domains give the asserted C^1 closeness. The transverse width has a uniform positive lower bound after the fixed margin losses. $\square$

8.4 Two cycles cannot give the same leaf

Lemma 8.5. *In the setting of Lemma 8.4, suppose two initial strips are obtained from independent primitive cycles of one central smooth torus. One can choose a finite required initial block length and a positive C^1 error tolerance from the central torus and its two cycles, before restricting the deformation parameter, with the following property. For strips meeting these bounds, if*

$$([\sigma_t], e) \neq 0, \tag{8.6}$$

then L holds at every point of the smaller tube covered by both initial strip charts.

Proof. Fix such a point x and take one leaf of each strip through it, using sheets whose preimages lie inside the buffered initial blocks. Identify a smooth tube of the central torus with a disc times that torus, and let q be its smooth projection to the torus. On a sufficiently large finite box in the universal plane cover of the torus, the lifted projections of these two leaf parametrizations are C^1 close to invertible affine maps. The prescribed cylinder lengths may be chosen to include this box and every lattice translate used below, with a fixed positive buffer. These choices are made on the central model; the required C^1 tolerance is then fixed on these finite buffered domains. A small C^1 perturbation of an invertible affine map is injective on a slightly smaller convex box, and its image contains a further smaller box. We may therefore invert the two projections and express the two leaf pieces as graphs over one common convex box Q in the torus cover. Choose this box to contain a fundamental parallelogram for the two cycles and buffers beyond every edge. The finite block lengths were chosen large enough to accommodate these translates and buffers.

Each graph is periodic under its own primitive cycle wherever both points and the required buffers belong to Q . To check this exact assertion, shifting w by 1 leaves its strip map unchanged. The lift of its torus projection changes by an integral lattice vector; closeness to the original chart

forces that vector to be exactly the chosen cycle. Inversion and uniqueness in the buffered box give the claimed graph equality.

Suppose the two holomorphic curve germs at x were equal. The set where the graph germs agree is open in Q . It is also closed. At a limit of such points, both graph images have the same limiting point. In small immersion charts at that point, the two embedded holomorphic curve germs have accumulating intersections, so the identity theorem makes them equal. The use of the smooth projection causes no difficulty here: it only supplies graph coordinates, while the identity theorem is applied in holomorphic surface charts. Connectedness of Q now makes the graphs agree throughout Q .

The common graph is periodic under both edge identifications. It consequently defines a compact smooth torus mapped into the tube whose local image is a holomorphic curve. Let m be the index of the sublattice generated by the two cycles in the full period lattice. Contraction of the disc component shows that the represented integral homology class in X_t is me , with $m > 0$; in the application from Lemma 8.3, $m = 1$. The pullback of σ_t to a holomorphic curve is zero, contradicting (8.6). The two leaf germs are therefore distinct.

They need not be transverse at x . Take a local foliation chart for the second strip. Its transverse coordinate restricts to a nonconstant holomorphic function on the first leaf, since the two germs are distinct. Its derivative is nonzero at points arbitrarily close to x . At such a point the first leaf meets a member of the second strip family transversely. Apply Remark 2.5 at this intersection, assigning the first strip, whose leaf contains x , to the role of σ_2 and the transverse partner to the role of σ_1 . Proposition 2.2 then gives L at x by the permitted displacement along the σ_2 leaf. $\square$

Proof of Theorem 8.1. By Lemma 8.3, all central fibers except finitely many have two independent safe cycle choices. Let $A \subset X_0$ be the union of the excluded fibers and the singular fibers. This is a proper analytic subset of dimension at most one.

We make a finite compact preparation before deforming. At every point of $X_0 \setminus A$, choose smaller regular torus patches with two safe cycles and larger buffered patches. At a point of A , choose a small holomorphic coordinate line disc through the point whose boundary avoids A . Such a disc exists because a generic line is not contained in A and meets it discretely. Two sufficiently small different tilts give transverse discs with boundaries still outside A . Their centers may vary in a smaller coordinate neighborhood. Compactness of X_0 permits a finite choice of these center neighborhoods and regular patches. Enlarge the compact regular data to include all the disc boundaries. Every selected safe action then has a positive common margin from its finite forbidden set.

Apply Lemma 8.4 to these finitely prepared data, with blocks long enough for Lemma 8.5. On all sufficiently small deformations in $\mathcal{D}_v$, the initial charts still cover fixed smaller tube patches, and the coordinate-disc boundaries remain in those patches. If (8.6) holds, Lemma 8.5 gives L at every point of the required regular patches. Corollary 2.10 gives L in the reserved center neighborhoods. Thus X_t has L everywhere.

Finally (8.6) holds on a nonempty relatively open subset arbitrarily close to P_0 in $\mathcal{D}_v$. Indeed $(e, v) = 0$, but the functional $(e, \cdot)$ does not vanish identically on the positive vectors of $v^\perp$. Varying the other normalized positive direction therefore breaks the equation $(e, [\sigma_t]) = 0$. Local Torelli identifies our small deformation neighborhood with a period neighborhood, giving the desired nonempty relatively open set $\mathcal{O}_v$. For any other marked K3 with the same period, global Torelli after matching Kähler chambers identifies the underlying surface, so the property has the stated period-theoretic scope. $\square$

9 Period dynamics and the conclusion

We now pass from the two open regions constructed above to every K3 surface. The distinction between no rational direction, one rational direction, and a rational plane is essential. In particular, density of an Oka locus alone would not justify this step. The corrected period-orbit analysis is due to Verbitsky [29]; the recent dense-period application in [30] uses the same broad strategy of geometric flexibility, arithmetic period dynamics, and Torelli theory. Here the two preceding geometric constructions supply open regions for both nonrational orbit types. We give the particular group argument needed for this transfer.

Write $V = \Lambda \otimes \mathbb{R}$, and let $G = \mathrm{SO}_0(V)$ be the identity component. Passing to finite-index subgroups, the integral isometry group gives an arithmetic lattice $\Gamma \subset G$ by the arithmetic lattice theorem [4].

Lemma 9.1. *Let $P \in \mathcal{D}$.*

1. *If $P \cap (\Lambda \otimes \mathbb{Q}) = \{0\}$, its integral-isometry orbit is dense in $\mathcal{D}$.*
2. *If $P \cap (\Lambda \otimes \mathbb{Q}) = \mathbb{Q}v$ for a primitive integral vector v , its orbit under integral isometries fixing v is dense in $\mathcal{D}_v$.*

Proof. We first work with an ordered orthonormal positive frame of P . Its connected pointwise stabilizer is $H = \mathrm{SO}_0(1, 19)$. This group is generated by one-parameter unipotent subgroups. Ratner's orbit closure theorem [27], applied to $H \cdot e\Gamma$ in G/Γ , gives

$$\overline{H \cdot e\Gamma} = S \cdot e\Gamma$$

for a connected closed subgroup $H \subset S \subset G$ such that $S \cap \Gamma$ is a lattice in S . Thus H is the pointwise stabilizer of the actual plane P throughout, and the lattice matrices in every intermediate group are rational in the original lattice coordinates.

Put $W = P^\perp$. As a module for $\mathfrak{h} = \mathfrak{so}(W)$, the orthogonal block-matrix decomposition is

$$\mathfrak{so}(V) = \mathfrak{h} \oplus \mathfrak{so}(P) \oplus (W \otimes P).$$

The standard real representation W is absolutely irreducible. Thus an intermediate Lie algebra obtains its mixed part by choosing a subspace $A \subset P$, with mixed module $W \otimes A$. If $\dim A = 2$, brackets give the full Lie algebra. If $\dim A = 1$, they give $\mathfrak{so}(W \oplus A)$, the stabilizer of the positive line $A^\perp \cap P$. Including the nonzero rotation algebra of P in this case generates the missing mixed summand and hence the full algebra. If $A = 0$, the only possibilities are $\mathfrak{h}$ and $\mathfrak{h} \oplus \mathfrak{so}(P)$.

The proper alternatives force rational directions in P . For the stabilizer of one positive line, Borel density [3] makes its lattice Zariski dense in that stabilizer. As the lattice consists of rational matrices, its Zariski closure is defined over $\mathbb{Q}$. Its fixed line is therefore rational. The same reasoning for H makes its fixed plane rational.

There is also no difficulty from the compact rotation factor in the remaining algebra. Up to finite isogeny the corresponding connected group is $\mathrm{SO}_0(W) \times \mathrm{SO}(P)$. Projection to the first factor is proper because its kernel is compact. Thus the image of the lattice meets each compact set in only finitely many points and is discrete. The finite invariant measure on the original quotient pushes forward to a finite invariant measure on the projected quotient, so the image is a lattice there. Borel density implies that the Lie algebra of the Zariski closure of the original lattice surjects onto $\mathfrak{h}$. Its derived algebra is exactly $\mathfrak{h}$: it is contained there because the other factor is commutative, and it maps onto $\mathfrak{h}$ because that algebra is simple. The Zariski closure and its

derived algebra are defined over $\mathbb{Q}$, so the fixed space of the derived algebra, namely P , is rational. Here the rationality of the Zariski closure of rational matrices follows directly from the vanishing equations: in each bounded degree their linear conditions have rational coefficients and hence a rational basis of solutions.

If P has no rational direction, none of these proper alternatives is possible. Ratner's closure is consequently the whole homogeneous space. Equivalently $H\Gamma$ is dense in G . Inversion makes ΓH dense as well, and projection to G/H makes the Γ -orbit of the chosen frame dense in frame space. The projection from frame space to $\mathcal{D}$ proves the first assertion. The use of the frame stabilizer H , rather than $H \times \mathrm{SO}(2)$, is what permits the application of Ratner's theorem.

For the second assertion, restrict to $v^\perp$, of signature $(2, 19)$. The integral stabilizer of v is arithmetic and gives a lattice in $G_v = \mathrm{SO}_0(v^\perp)$. More explicitly, the subgroup of integral isometries of the orthogonal lattice acting trivially on its discriminant group extends to the full K3 lattice while fixing v [24, Chapter 14, Section 2], so the integral stabilizer has finite index in the relevant orthogonal arithmetic group.

An oriented plane containing v is equivalent to a positive unit vector $u \in v^\perp$: choose the sign of u using the given plane orientation. The connected stabilizer of u is again $H = \mathrm{SO}_0(1, 19)$. This time

$$\mathfrak{so}(2, 19) = \mathfrak{h} \oplus W$$

as an $\mathfrak{h}$ -module, so irreducibility shows that H is maximal at Lie algebra level. Ratner's alternatives are a closed H -orbit or a dense orbit. In the closed case Borel density makes the line $\mathbb{R}u$ rational in the rational quadratic space $v^\perp$. This would give a second rational direction in P , contrary to the hypothesis. The orbit is therefore dense. The connected group G_v is transitive on positive unit vectors, so the resulting density is throughout $\mathcal{D}_v$. $\square$

Proof of Theorem 1.1. Choose a marking of X and let $P \in \mathcal{D}$ be its period. The rational dimension of $P \cap (\Lambda \otimes \mathbb{Q})$ is zero, one, or two. We show in each case that X has L everywhere.

If it is zero, Lemma 9.1 makes the marked period orbit dense in $\mathcal{D}$. It meets the nonempty open subset provided by Proposition 7.1. Re-marking does not change the complex surface. Moreover K3 surfaces with the same marked period are isomorphic after their markings are forgotten: signs and reflections in algebraic roots match their Kähler chambers, and global Torelli then supplies an isomorphism. Thus the all-point property L in that open region holds on X .

If the rational dimension is one, let v be a primitive integral generator of that direction. It is positive because P is positive. The second part of Lemma 9.1 allows us to re-mark within the stabilizer of v so that the period lies in the relatively open set $\mathcal{O}_v$ of Theorem 8.1. The same Torelli argument gives L everywhere on X . Notice that this step uses an open subset inside the fixed- v locus itself.

If the rational dimension is two, P is a rational plane. Its rational orthogonal complement has signature $(1, 19)$. The positive cone is open and nonempty, so it contains a rational vector, which can be multiplied by an integer to give a positive integral $(1, 1)$ -class. The K3 projectivity criterion implies that X is projective. Proposition 5.1 therefore gives L at every point of X .

Theorem 3.1 now applies to X , proving the stated approximation statement with all its quantifiers. The estimates there are compact-uniform estimates for maps into X . On the compact target, any two smooth Hermitian distances are uniformly equivalent, so the conclusion holds for the arbitrary metric specified in the statement. $\square$

10 Interpolation and further consequences

10.1 Interpolation and dense immersions

We now obtain approximation and interpolation from open Riemann surfaces and derive the dense entire immersion in Corollary 1.2. An *open Riemann surface* is a connected noncompact Riemann surface. A compact subset is *Runge* if its complement has no relatively compact component. The notation j_a^k records the value and derivatives through order k in local coordinates, and d_S denotes any fixed distance inducing the topology of S .

Corollary 10.1. *Let S be a complex K3 surface, R an open Riemann surface, $K \subset R$ a compact Runge set, and $A = \{a_j : j \geq 1\} \subset R$ a closed discrete set of distinct points. Let $k_j \geq 1$ be integers. For every continuous map $h : R \rightarrow S$ holomorphic near $K \cup A$ and every $\epsilon > 0$, there is a holomorphic map $f : R \rightarrow S$, homotopic to h , satisfying*

$$\sup_{z \in K} d_S(f(z), h(z)) < \epsilon, \quad j_{a_j}^{k_j} f = j_{a_j}^{k_j} h \quad (j \geq 1).$$

Finite or empty interpolation sets are also allowed. If each prescribed first derivative is nonzero, f can be chosen to be an immersion.

Proof. Theorem 1.1 and the convex approximation characterization [13, Theorem 0.1] imply that S is Oka. Every Oka manifold is Oka-1, which is precisely the approximation and interpolation property stated here. The CAP-to-interpolation implication is established in [12, Corollary 1.3]; for the Oka-1 formulation with individually prescribed jet orders, see [1, Definition 1.1 and the following discussion]. Since $\dim_{\mathbb{C}} S = 2$, the same interpolating map can be chosen to be an immersion by [1, Corollary 2.10] when all prescribed first derivatives are nonzero. $\square$

Proof of Corollary 1.2. Choose a countable dense sequence $(x_j)_{j \geq 1}$ in the ordinary topology of S , set $x_0 = x$, and choose nonzero vectors $v_j \in T_{x_j} S$, with $v_0 = v$. At the distinct source points $a_j = 3j$, choose holomorphic germs $g_j : (\mathbb{C}, a_j) \rightarrow (S, x_j)$ with $g'_j(a_j) = v_j$. Explicitly, if κ_j is a chart taking x_j to the origin in a ball in $\mathbb{C}^2$, put

$$g_j(z) = \kappa_j^{-1}((z - a_j)d\kappa_j(v_j)).$$

Choose $0 < r_j < 1/3$ so that this formula is defined near $\{|z - a_j| \leq 3r_j\}$. These closed discs are disjoint and locally finite.

The germs are restrictions of a single continuous map $h : \mathbb{C} \rightarrow S$. To construct it, fix $b \in S$ and a continuous path from each x_j to b . Use g_j on $|z - a_j| \leq r_j$. On the annulus $r_j \leq |z - a_j| \leq 2r_j$, multiply $\kappa_j(g_j(z))$ by a continuous radial cutoff equal to one near its inner boundary and zero near its outer boundary. Convexity of the coordinate ball keeps the resulting map in the chart. On $2r_j \leq |z - a_j| \leq 3r_j$, follow the chosen path from x_j to b with a radial parameter constant near the two boundaries. Put $h = b$ elsewhere. Local finiteness and the boundary agreements prove continuity; h is holomorphic near every a_j and near the compact Runge disc $K = \{|z| \leq r_0/2\}$.

Apply Corollary 10.1 to h , K , and the closed discrete sequence $(a_j)_{j \geq 0}$, prescribing the first jet at every a_j . Since every v_j is nonzero, this gives an interpolating holomorphic immersion $f : \mathbb{C} \rightarrow S$. Thus

$$f(a_j) = x_j, \quad f'(a_j) = v_j \quad (j \geq 0).$$

The case $j = 0$ gives the prescribed point and exact tangent vector. The image contains the dense sequence $(x_j)_{j \geq 1}$, proving ordinary density. If S is projective, its Zariski closed subsets are closed in the ordinary complex topology, proving the last assertion in that case. $\square$

Strong dominability. For every complex K3 surface S and every $x \in S$, there is a holomorphic map $F_x : \mathbb{C}^2 \rightarrow S$ with $F_x(0) = x$ and $d(F_x)_0 : \mathbb{C}^2 \rightarrow T_x S$ an isomorphism. This is strong dominability by $\mathbb{C}^2$ in the sense of [16, Definition 1]. Indeed, choose a local coordinate inverse at $0 \in \mathbb{C}^2$ taking 0 to x , and extend it continuously to $\mathbb{C}^2$ by a radial cutoff inside the coordinate ball, keeping it unchanged near 0 . The Oka property with first-order jet interpolation at $\{0\}$ [12, Proposition 1.2 and Corollary 1.3] gives an entire map with the same value and differential at 0 .

10.2 Surface classification and global sprays

The K3 theorem also settles the Oka property for Enriques surfaces. Xie and Zhao proved it for unnodal Enriques surfaces, namely those containing no (-2) -curves [30, Theorem C and Corollary 3.5]. The following consequence includes the nodal case. Its class-VII assertion has a separate input: the global spherical shell theorem [26, Theorem 1.1], which is not used in the K3 proof.

Corollary 10.2 (Oka surfaces in Kodaira dimension zero and class VII). *Every complex Enriques surface is Oka. More generally, every connected minimal compact complex surface of Kodaira dimension zero is Oka. A connected minimal compact complex surface of class VII is Oka if and only if it is a Hopf surface or an Enoki surface.*

Proof. Every complex Enriques surface has an unramified holomorphic K3 double cover [18, Introduction]. Its covering surface is Oka by Theorem 1.1, and the Oka property descends through holomorphic covering maps [16, Introduction]. This proves the Enriques assertion. Apart from K3 and Enriques surfaces, the minimal compact complex surfaces of Kodaira dimension zero are complex tori, bielliptic surfaces, and Kodaira surfaces, which are Oka by the established cases recalled in [16, Introduction].

A minimal class-VII surface has $b_1 = 1$ and $\kappa = -\infty$. When $b_2 > 0$, the global spherical shell theorem [26, Theorem 1.1] therefore applies, so the surface is a Kato surface. The class-VII decomposition recalled in [16, Introduction] now makes the list exhaustive: Hopf and Inoue surfaces for $b_2 = 0$, and Enoki, Inoue–Hirzebruch, and intermediate surfaces for $b_2 > 0$. Theorem 4 of [16] shows that exactly the Hopf and Enoki cases are Oka. $\square$

The classification assertion is restricted to minimal surfaces; no claim is made here for arbitrary blowups or for properly elliptic surfaces.

A complex manifold Y is *elliptic* in Gromov’s sense if it has a global dominating holomorphic spray: a holomorphic vector bundle $E \rightarrow Y$ and a holomorphic map $s : E \rightarrow Y$ such that $s(0_y) = y$ and the fiber differential $d(s|_{E_y})_{0_y} : E_y \rightarrow T_y Y$ is surjective for every $y \in Y$ [21].

Corollary 10.3 (Global holomorphic sprays). *Every projective complex K3 surface and every complex Enriques surface is elliptic in Gromov’s sense.*

Proof. Projective complex K3 surfaces are Oka by Theorem 1.1. Complex Enriques surfaces are projective [18, Section 2] and are Oka by Corollary 10.2. The conclusion follows from Forstnerič and Lárusson’s theorem that every projective Oka manifold is elliptic [17, Theorem 1.1]. $\square$

This last implication requires projectivity; no assertion about a global dominating spray on a nonprojective K3 surface is made here.

References

- [1] Antonio Alarcón and Franc Forstnerič. Oka-1 manifolds. *Mathematische Zeitschrift*, 311, 2025. Paper No. 33.
- [2] Michela Artebani and Alessandra Sarti. Non-symplectic automorphisms of order 3 on K3 surfaces. *Mathematische Annalen*, 342:903–921, 2008.
- [3] Armand Borel. Density properties for certain subgroups of semi-simple groups without compact components. *Annals of Mathematics*, 72(1):179–188, 1960.
- [4] Armand Borel and Harish-Chandra. Arithmetic subgroups of algebraic groups. *Annals of Mathematics*, 75(3):485–535, 1962.
- [5] Gregory T. Buzzard and Steven S. Y. Lu. Algebraic surfaces holomorphically dominable by $\mathbb{C}^2$. *Inventiones Mathematicae*, 139(3):617–659, 2000.
- [6] Benoît Cadorel, Frédéric Campana, and Erwan Rousseau. Hyperbolicity and specialness of symmetric powers. *Journal de l’École polytechnique—Mathématiques*, 9:381–430, 2022.
- [7] Frédéric Campana. Orbifolds, special varieties and classification theory. *Annales de l’Institut Fourier*, 54(3):499–630, 2004.
- [8] Gui-Qiang G. Chen and Siran Li. Global weak rigidity of the Gauss–Codazzi–Ricci equations and isometric immersions of Riemannian manifolds with lower regularity, 2017. arXiv:1607.06862v2.
- [9] Xi Chen and Frank Gounelas. Curves of maximal moduli on K3 surfaces. *Forum of Mathematics, Sigma*, 10, 2022. Paper No. e36.
- [10] Xi Chen, Frank Gounelas, and Christian Liedtke. Curves on K3 surfaces. *Duke Mathematical Journal*, 171(16):3283–3362, 2022.
- [11] Franc Forstnerič. Noncritical holomorphic functions on Stein manifolds. *Acta Mathematica*, 191:143–189, 2003.
- [12] Franc Forstnerič. Extending holomorphic mappings from subvarieties in Stein manifolds. *Annales de l’Institut Fourier*, 55(3):733–751, 2005.
- [13] Franc Forstnerič. Runge approximation on convex sets implies the Oka property. *Annals of Mathematics*, 163(2):689–707, 2006.
- [14] Franc Forstnerič. *Stein Manifolds and Holomorphic Mappings: The Homotopy Principle in Complex Analysis*, volume 56 of *Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge*. Springer, second edition, 2017.
- [15] Franc Forstnerič and Finnur Lárusson. Survey of Oka theory. *New York Journal of Mathematics*, 17a:11–38, 2011.
- [16] Franc Forstnerič and Finnur Lárusson. Holomorphic flexibility properties of compact complex surfaces. *International Mathematics Research Notices*, 2014(13):3714–3734, 2014. arXiv:1207.4838v3.

- [17] Franc Forstnerič and Finnur Lárusson. Every projective Oka manifold is elliptic. *Mathematical Research Letters*, 33(1):297–312, 2026. arXiv:2502.20028v6, revised May 5, 2026. [Manuscript PDF](#).
- [18] Francisco Javier Gallego, Miguel González, and Bangere P. Purnaprajna. K3 double structures on Enriques surfaces and their smoothings, 2006. arXiv:math/0604629v2.
- [19] Hans Grauert. Ein Theorem der analytischen Garbentheorie und die Modulräume komplexer Strukturen. *Publications Mathématiques de l’IHÉS*, 5:5–64, 1960.
- [20] Valeri Gritsenko, Klaus Hulek, and Gregory K. Sankaran. Abelianisation of orthogonal groups and the fundamental group of modular varieties. *Journal of Algebra*, 322(2):463–478, 2009. Proposition 3.3(i).
- [21] Mikhael Gromov. Oka’s principle for holomorphic sections of elliptic bundles. *Journal of the American Mathematical Society*, 2(4):851–897, 1989.
- [22] Lars Hörmander. L^2 estimates and existence theorems for the $\bar{\partial}$ operator. *Acta Mathematica*, 113:89–152, 1965.
- [23] Lars Hörmander. *An Introduction to Complex Analysis in Several Variables*. North-Holland, third edition, 1990.
- [24] Daniel Huybrechts. *Lectures on K3 Surfaces*, volume 158 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2016.
- [25] Ljudmila Kamenova, Steven Lu, and Misha Verbitsky. Kobayashi pseudometric on hyperkähler manifolds. *Journal of the London Mathematical Society*, 90(2):436–450, 2014.
- [26] OpenAI. Global Spherical Shells on Minimal Surfaces of Class VII. OpenAI Math Release preprint [OAI:Global-Spherical-Shells-on-Minimal-Surfaces-of-Class-VII-September-24-2026](#), 2026.
- [27] Marina Ratner. Raghunathan’s topological conjecture and distributions of unipotent flows. *Duke Mathematical Journal*, 63(1):235–280, 1991.
- [28] Yum-Tong Siu. Every Stein subvariety admits a Stein neighborhood. *Inventiones Mathematicae*, 38:89–100, 1976.
- [29] Misha Verbitsky. Ergodic complex structures on hyperkähler manifolds: an erratum, 2017. arXiv:1708.05802v1.
- [30] Song-Yan Xie and Shengyuan Zhao. Periods of Oka K3 surfaces are dense, 2026. arXiv:2608.24392v2, revised September 17, 2026.